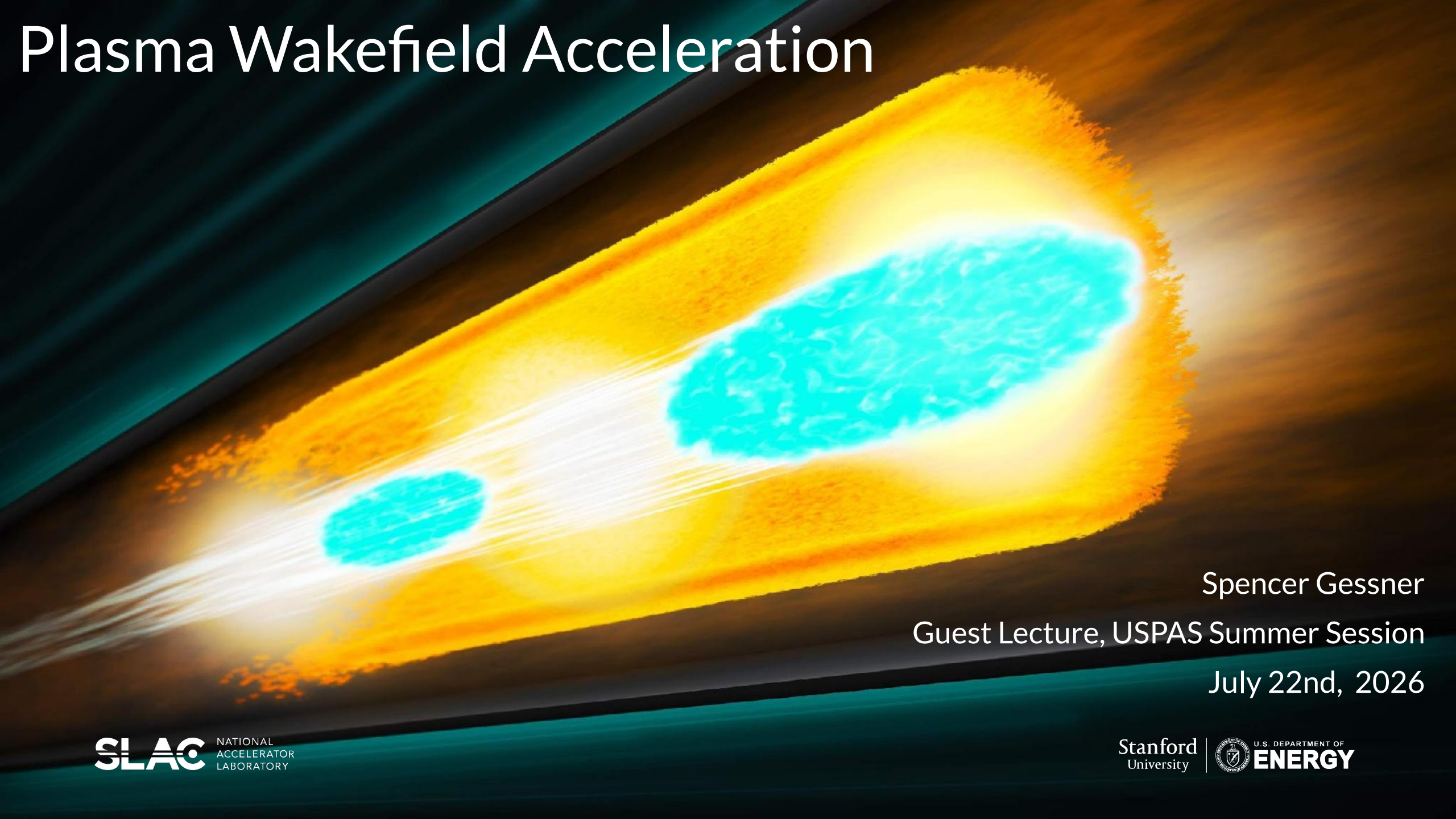


Plasma Wakefield Acceleration



Spencer Gessner

Guest Lecture, USPAS Summer Session

July 22nd, 2026

Reference Material

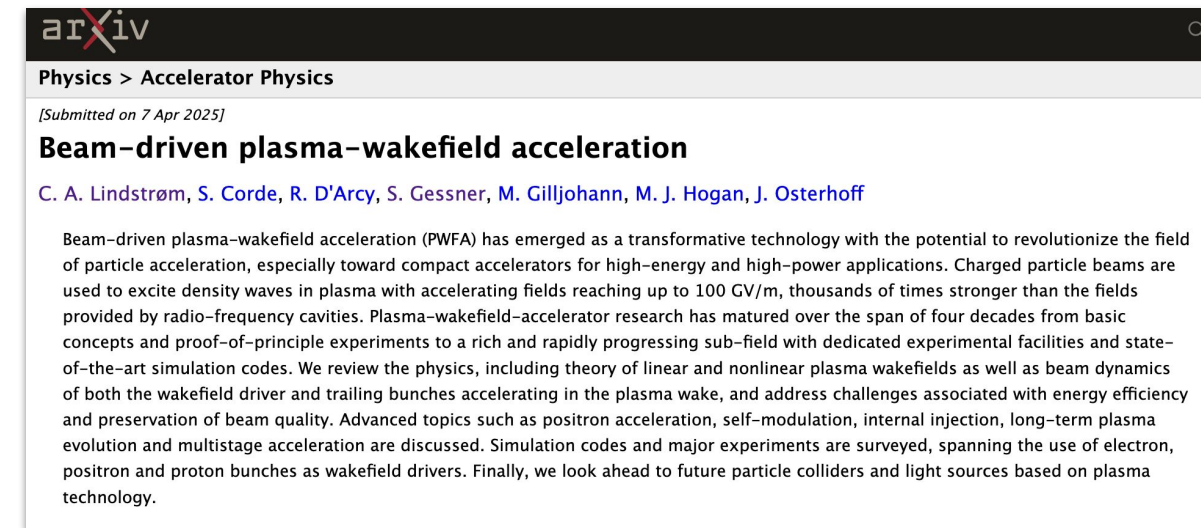
Mike Litos (CU Boulder) and I teach an USPAS course titled “Particle-Driven Plasma Wakefield Accelerators”. The most recent class was last summer.

Two recent review articles:

- “Beam-driven plasma wakefield acceleration” C. Lindstrom et al (under review for RMP) [link](#)
- “Positron acceleration in plasma wakefields” G. Cao et al, PRAB 2024 [link](#)



<https://sites.google.com/view/uspas-plasma-wakefield-2025/>



<https://arxiv.org/abs/2504.05558>

Outline

What's so exciting about Plasma Wakefield Acceleration?

Connections between beam and plasma physics

Derivation of plasma wakefields in the linear regime

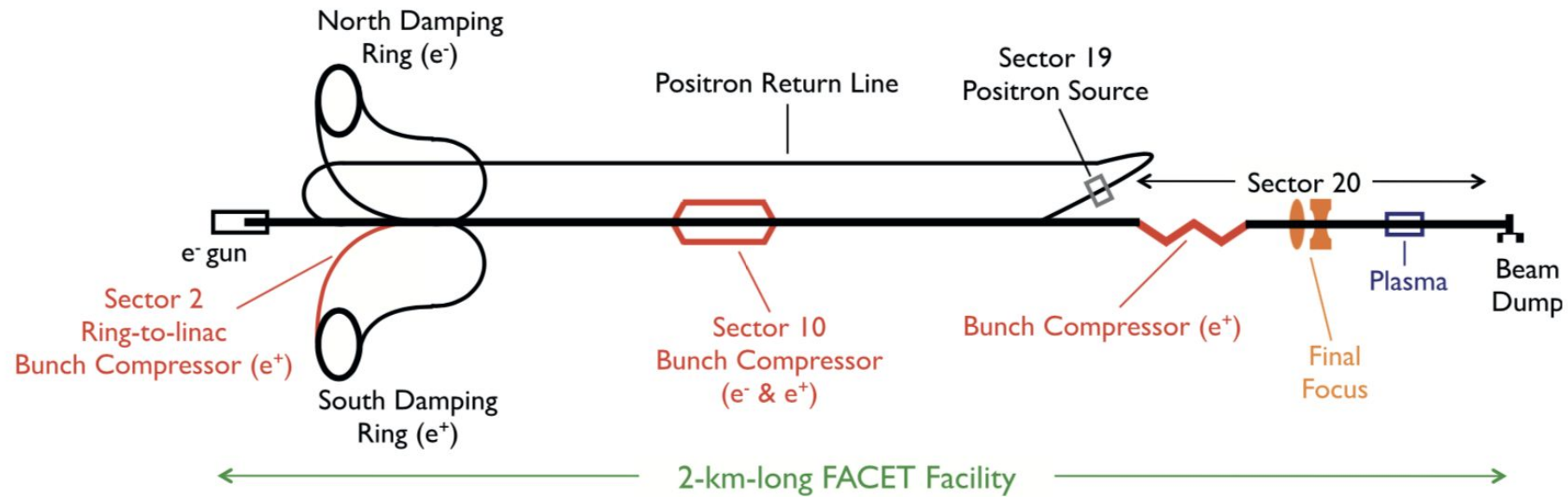
Ideal conditions for plasma acceleration

Future colliders and the positron problem

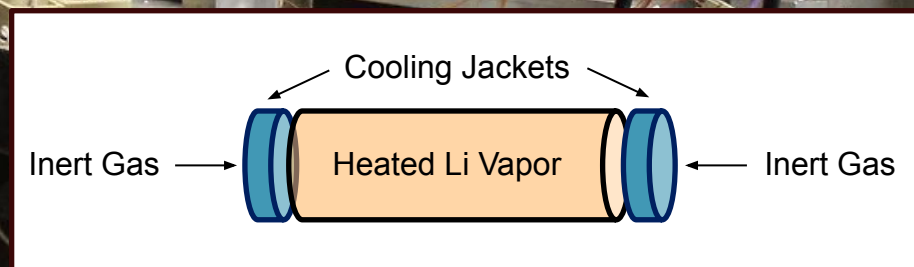
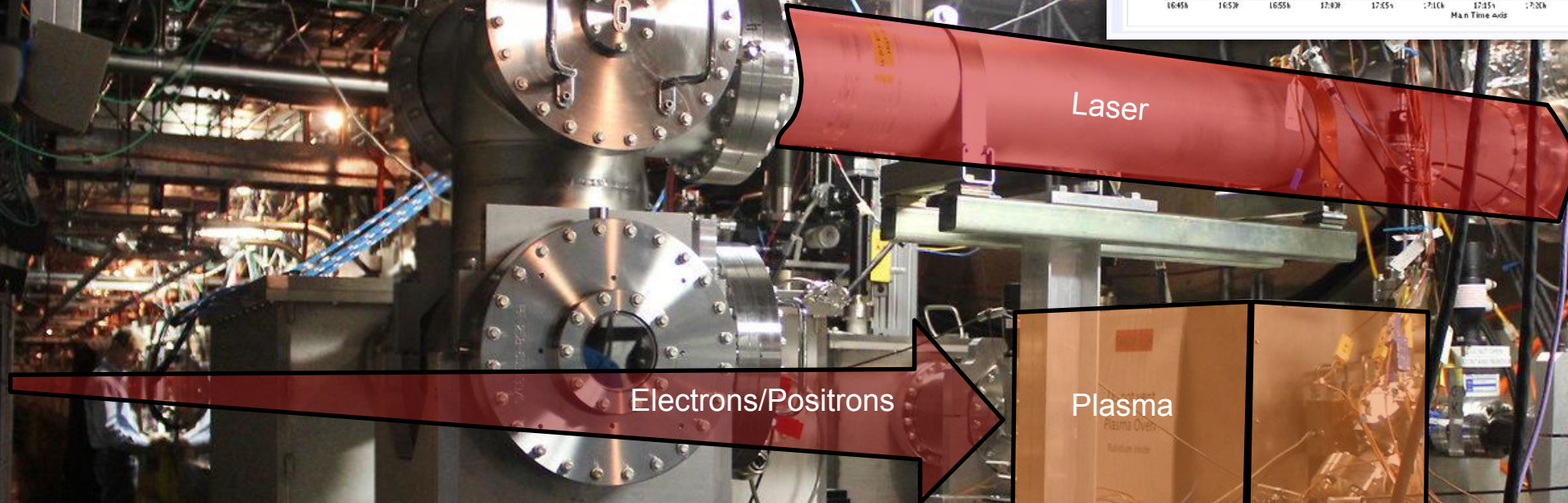
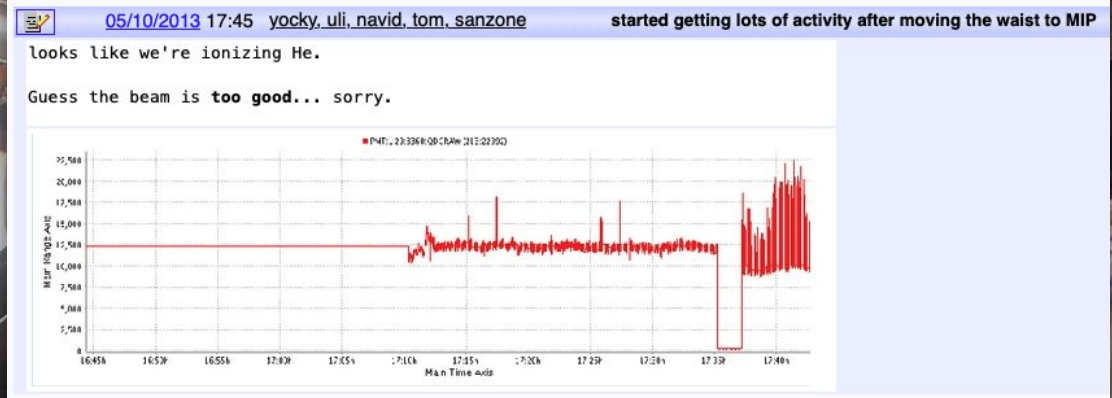
R&D at FACET

FACET (2011-2016)

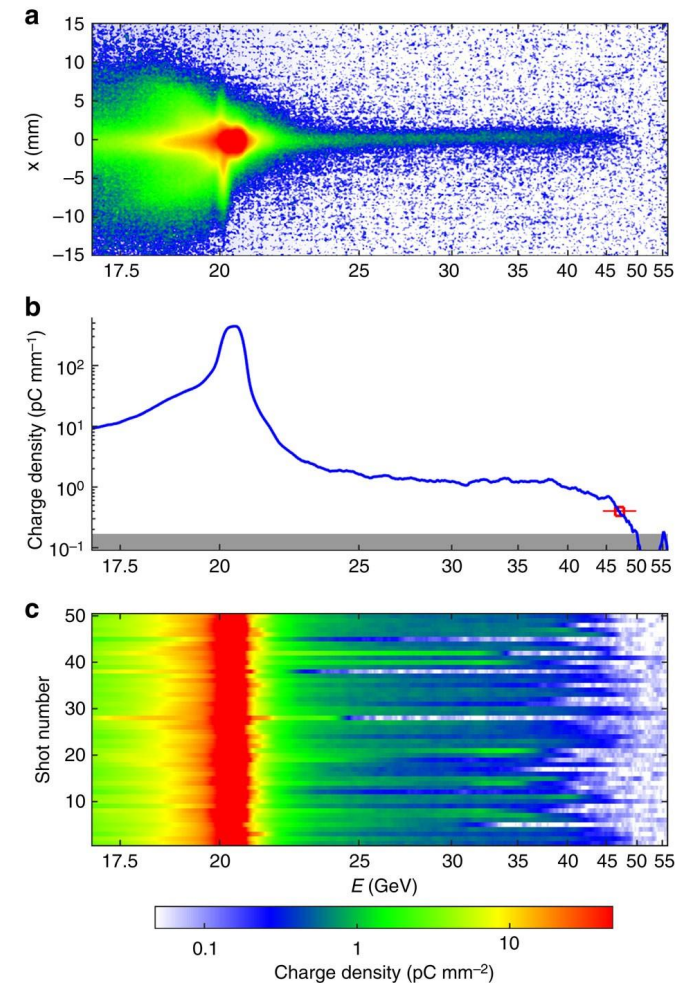
SLAC



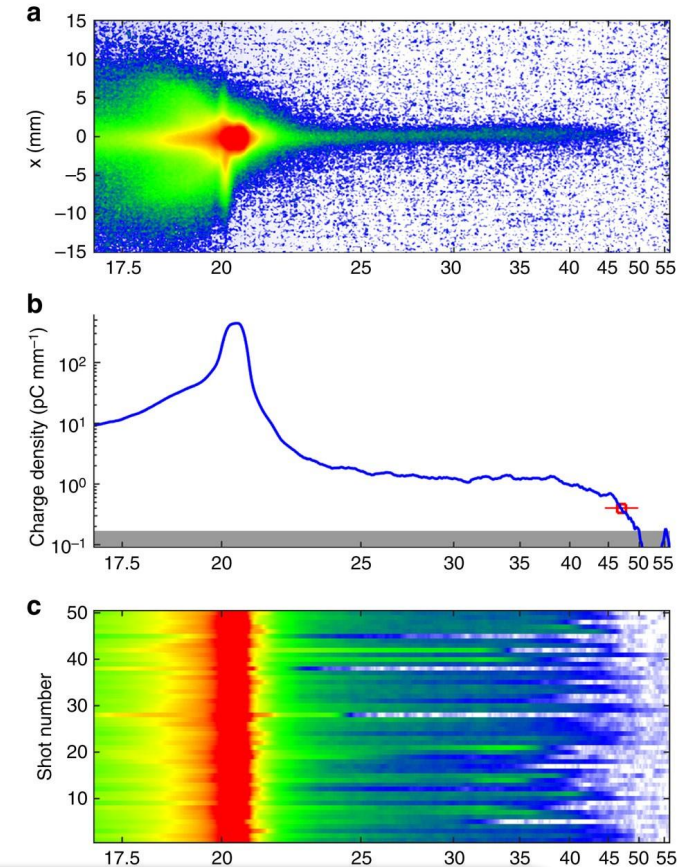
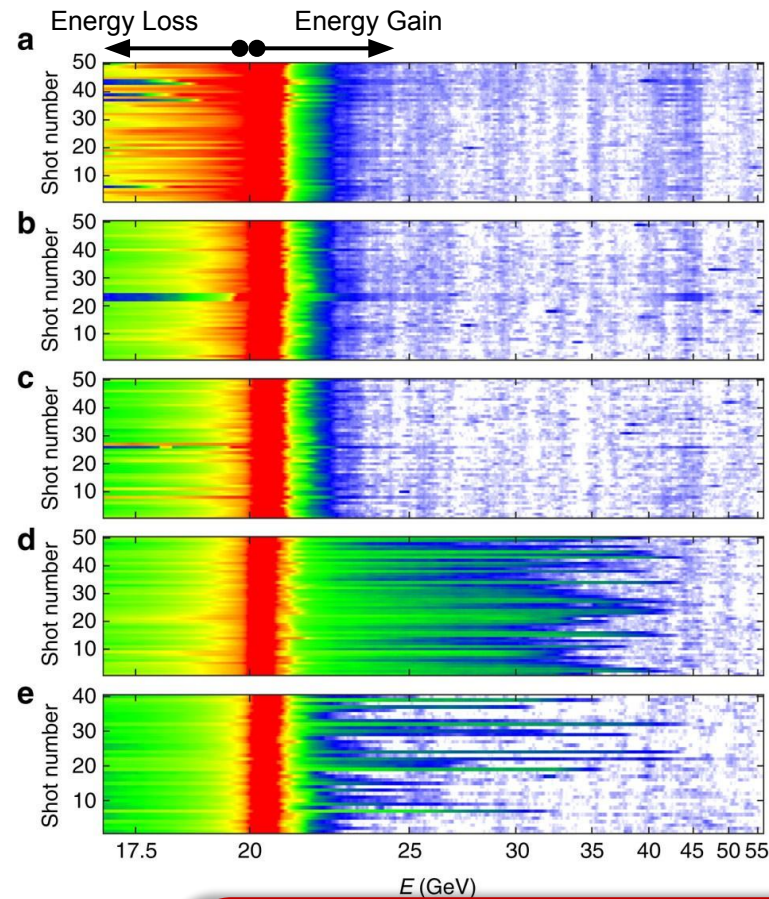
FACET



Ultra High Field Plasma Acceleration

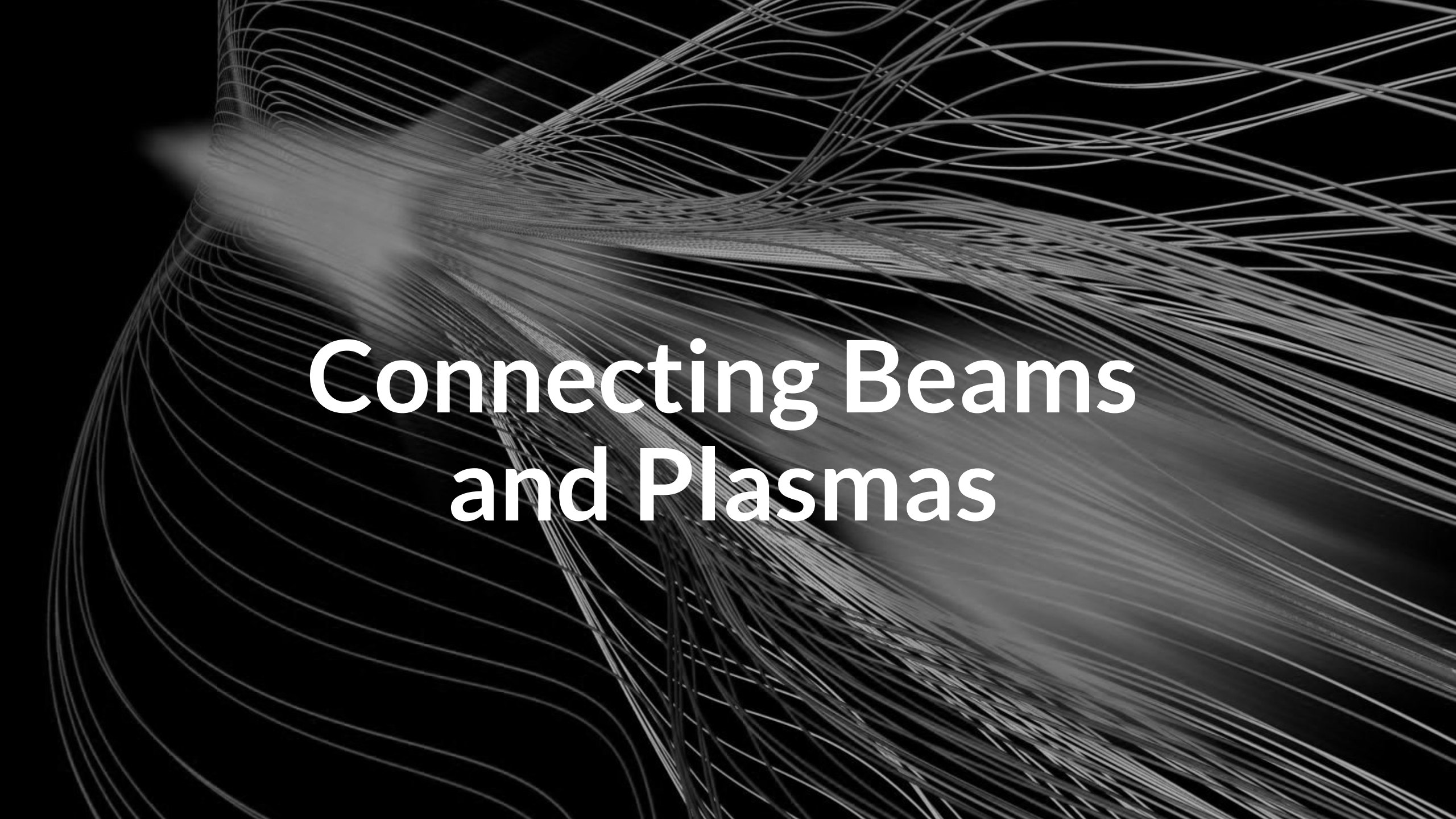


Ultra High Field Plasma Acceleration



We more than doubled the energy of some particles in the beam.

The accelerating field reached **150 GV/m**, about 7500x larger than the field in the SLAC linac RF cavities.



Connecting Beams and Plasmas

Distribution Functions

Beams and plasmas are both described by particle distribution functions.

$$f(\mathbf{x}, \mathbf{p}, t)$$
$$n(\mathbf{x}, t) = \int f(\mathbf{x}, \mathbf{p}, t) d\mathbf{p}$$

The main differences between beams and plasmas are:

- particle beams are **non-neutral**
- particle beams are **relativistic**

The Vlasov Equation

The Vlasov equation describes the evolution of the distribution function under the influence of conservative (non-dissipative) forces:

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\mathbf{p}}{\gamma m} \cdot \nabla f + \mathbf{F} \cdot \nabla_p f = 0$$

The Vlasov Equation

The Vlasov equation describes the evolution of the distribution function under the influence of conservative (non-dissipative) forces:

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\mathbf{p}}{\gamma m} \cdot \nabla f + \mathbf{F} \cdot \nabla_p f = 0$$

We can derive the continuity equation and fluid equation of motion by taking “moments” of the Vlasov equation:

$$\frac{\partial n}{\partial t} + \nabla \cdot (n\mathbf{v}) = 0$$

$$\left[\frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla \right] \mathbf{p} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

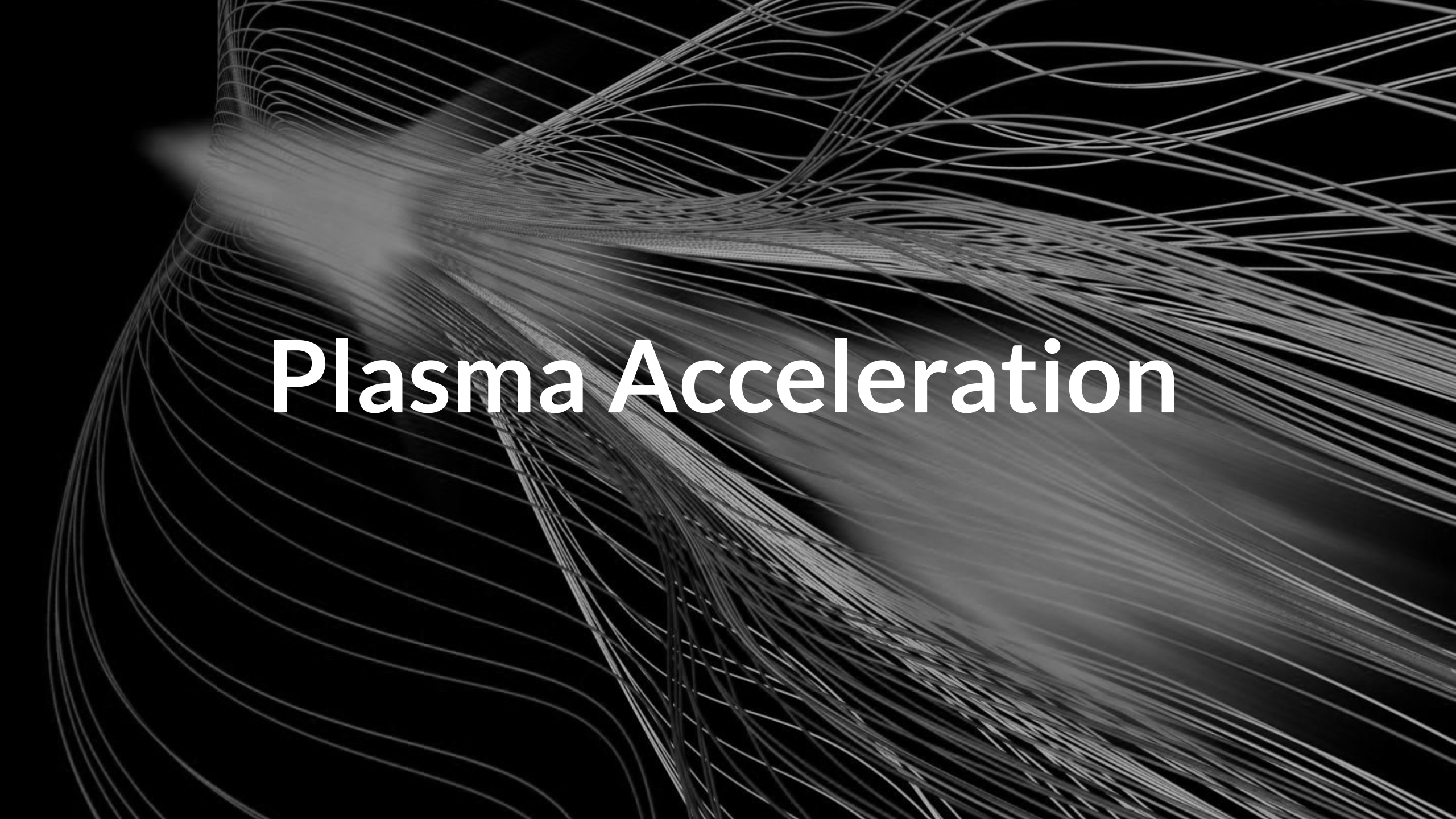
These are the equations we will need to describe the evolution of the plasma.

The Vlasov Equation

It's even possible to derive the beam envelope equation directly from Vlasov equation:

$$\frac{d^2\sigma_x}{dz^2} = -k_\beta^2\sigma_x + \frac{\varepsilon_g^2}{\sigma_x^3}$$

but we won't do that here...

The background is a black field filled with a dense, complex network of thin, white, curved lines. These lines originate from the left side and fan out towards the right, creating a sense of motion and expansion. The lines are not straight, but rather follow a curved, almost helical path, which could represent the trajectories of particles or the structure of a plasma jet. The overall effect is one of dynamic energy and acceleration.

Plasma Acceleration

Plasma Acceleration in the Linear Regime

Plasmas are **electromagnetic fluids**.

Maxwell's Equations

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0},$$

$$\nabla \cdot \vec{B} = 0,$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t},$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

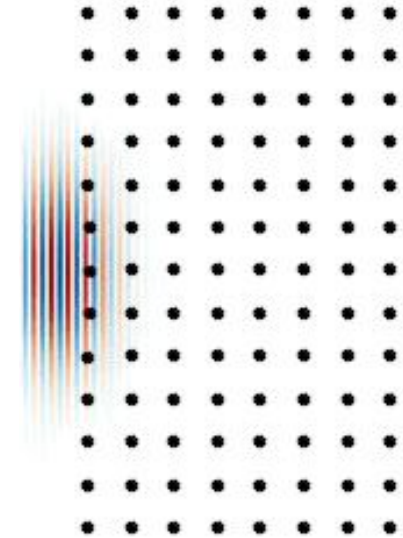
Continuity Equation

$$\frac{\partial n}{\partial t} + \nabla \cdot (n\vec{v}) = 0$$

Lorentz Force Law for Fluids

$$mn \left[\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v} \right] = qn(\vec{E} + \vec{v} \times \vec{B})$$

These equations are for *cold* plasmas: $T=0, P=0$



<https://wp.lancs.ac.uk/spiral/research/laser-driven-plasma-wakefield-acceleration/>

Plasma Acceleration in the Linear Regime

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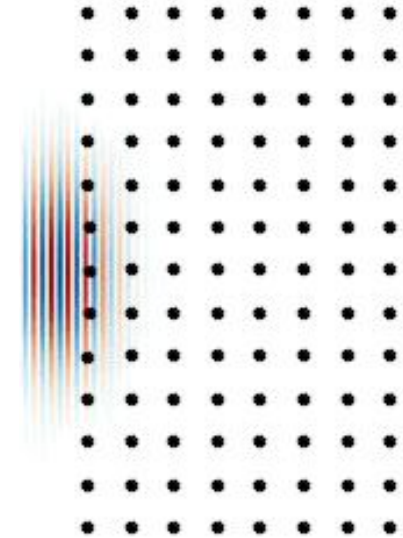
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These equations are for *cold* plasmas: $T=0, P=0$



<https://wp.lancs.ac.uk/spiral/research/laser-driven-plasma-wakefield-acceleration/>

Our goal: find the response of the plasma to a *perturbing* particle beam
by *linearizing* these equations.

Step 1: Quantities of Interest

Plasmas are **electromagnetic fluids**.

Maxwell's Equations

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0},$$

$$\nabla \cdot \vec{B} = 0,$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t},$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Continuity Equation

$$\frac{\partial n}{\partial t} + \nabla \cdot (n\vec{v}) = 0$$

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$$mn \left[\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v} \right] = qn(\vec{E} + \vec{v} \times \vec{B})$$

Perturbative Expansion for Quantities of Interest

$$n = n_0 + n_1 + n_2 + \dots$$

$$\vec{v} = \vec{v}_0 + \vec{v}_1 + \vec{v}_2 + \dots$$

$$\vec{E} = \vec{E}_0 + \vec{E}_1 + \vec{E}_2 + \dots$$

$$\vec{B} = \vec{B}_0 + \vec{B}_1 + \vec{B}_2 + \dots$$

Step 1: Quantities of Interest

Plasmas are **electromagnetic fluids**.

Maxwell's Equations

$$\begin{aligned}\nabla \cdot \vec{E} &= \frac{\rho}{\epsilon_0}, \\ \nabla \cdot \vec{B} &= 0, \\ \nabla \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t}, \\ \nabla \times \vec{B} &= \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}\end{aligned}$$

Continuity Equation

$$\frac{\partial n}{\partial t} + \nabla \cdot (n\vec{v}) = 0$$

Lorentz Force Law for Fluids

$$mn \left[\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v} \right] = qn(\vec{E} + \vec{v} \times \vec{B})$$

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$$\vec{E} = \vec{E}_0 + \vec{E}_1 + \vec{E}_2 + \dots$$

$$\vec{B} = \vec{B}_0 + \vec{B}_1 + \vec{B}_2 + \dots$$

Quantities ϱ and J are proportional to n and v .

0th-order quantities are constant.

Higher-order quantities vary in space and time.

Step 1: Quantities of Interest

Plasmas are **electromagnetic fluids**.

Maxwell's Equations

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0},$$

$$\nabla \cdot \vec{B} = 0,$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t},$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Continuity Equation

$$\frac{\partial n}{\partial t} + \nabla \cdot (n\vec{v}) = 0$$

Lorentz Force Law for Fluids

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Perturbative Expansion for Quantities of Interest

$$n = n_0 + n_1 + n_2 + \dots$$

$$\vec{v} = \cancel{\vec{v}_0} + \vec{v}_1 + \vec{v}_2 + \dots$$

$$\vec{E} = \cancel{\vec{E}_0} + \vec{E}_1 + \vec{E}_2 + \dots$$

$$\vec{B} = \cancel{\vec{B}_0} + \vec{B}_1 + \vec{B}_2 + \dots$$

No initial plasma electron velocity. No background E field.

No background B field.

Step 2: Linearized Continuity Equation

Assume a perturbation in the plasma (electron) density:

$$n = n_0 + n_1 \quad \frac{n_1}{n_0} \lesssim 0.1$$

The plasma electrons start to move in response, but not too fast:

$$\vec{v} = \vec{v}_0 + \vec{v}_1 \quad \vec{v}_0 = 0 \quad v_1 \ll c$$

Continuity Equation (only keep 1st-order terms):

$$\frac{\partial n}{\partial t} + \nabla \cdot (n\vec{v}) = 0 \quad \Rightarrow \quad \frac{\partial n_1}{\partial t} + n_0 \nabla \cdot \vec{v}_1 = 0.$$

Step 3: Linearized Lorentz Force Equation

Assume the perturbation causes a small E, B fields:

$$\vec{E} = \vec{E}_0 + \vec{E}_1 \quad \vec{E}_0 = 0 \quad \vec{B} = \vec{B}_0 + \vec{B}_1 \quad \vec{B}_0 = 0$$

Lorentz Force Law for Fluids (only keep 1st-order terms):

$$mn \left[\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v} \right] = qn(\vec{E} + \vec{v} \times \vec{B}) \quad \Rightarrow \quad mn_0 \frac{\partial \vec{v}_1}{\partial t} = -en_0 \vec{E}_1$$

Step 4: Linearized Gauss's Law

Continuity equation relates n to v . Lorentz Force Law relates v to E .

Gauss's Law provides a relation from E to n .

Gauss's Law:

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \longrightarrow \frac{\rho_p + \rho_b}{\epsilon_0} = -\frac{e}{\epsilon_0}(n_1 + n_b) \quad \Rightarrow \quad \nabla \cdot \vec{E}_1 = \frac{-e(n_1 + n_b)}{\epsilon_0}$$

Step 4: Linearized Gauss's Law

Continuity equation relates n to v . Lorentz Force Law relates v to E .

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Gauss's Law:

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$$\frac{\partial n_1}{\partial t} + n_0 \nabla \cdot \vec{v}_1 = 0 \quad mn_0 \frac{\partial \vec{v}_1}{\partial t} = -en_0 \vec{E}_1 \quad \nabla \cdot \vec{E}_1 = \frac{-e(n_1 + n_b)}{\epsilon_0}$$

Step 5: Combine

Whisk it together

$$\rightarrow \frac{\partial^2 n_1}{\partial t^2} = \underbrace{-\omega_p^2 n_1}_{\text{Cold Langmuir wave equation}} \underbrace{- \omega_p^2 n_b}_{\text{extra bit from the e-beam}}$$

Cold Langmuir wave equation with an extra bit from the e-beam

$$\omega_p = \sqrt{\frac{n_0 e^2}{\epsilon_0 m}}$$

Co-Moving Coordinates

- Change coordinates for convenience:

$$\xi = z - ct$$

- Longitudinal derivative: $\frac{\partial}{\partial z} = \frac{\partial \xi}{\partial z} \frac{\partial}{\partial \xi} = \frac{\partial}{\partial \xi}$

- Time derivative: $\frac{\partial}{\partial t} = \frac{\partial \xi}{\partial t} \frac{\partial}{\partial \xi} = -c \frac{\partial}{\partial \xi}$

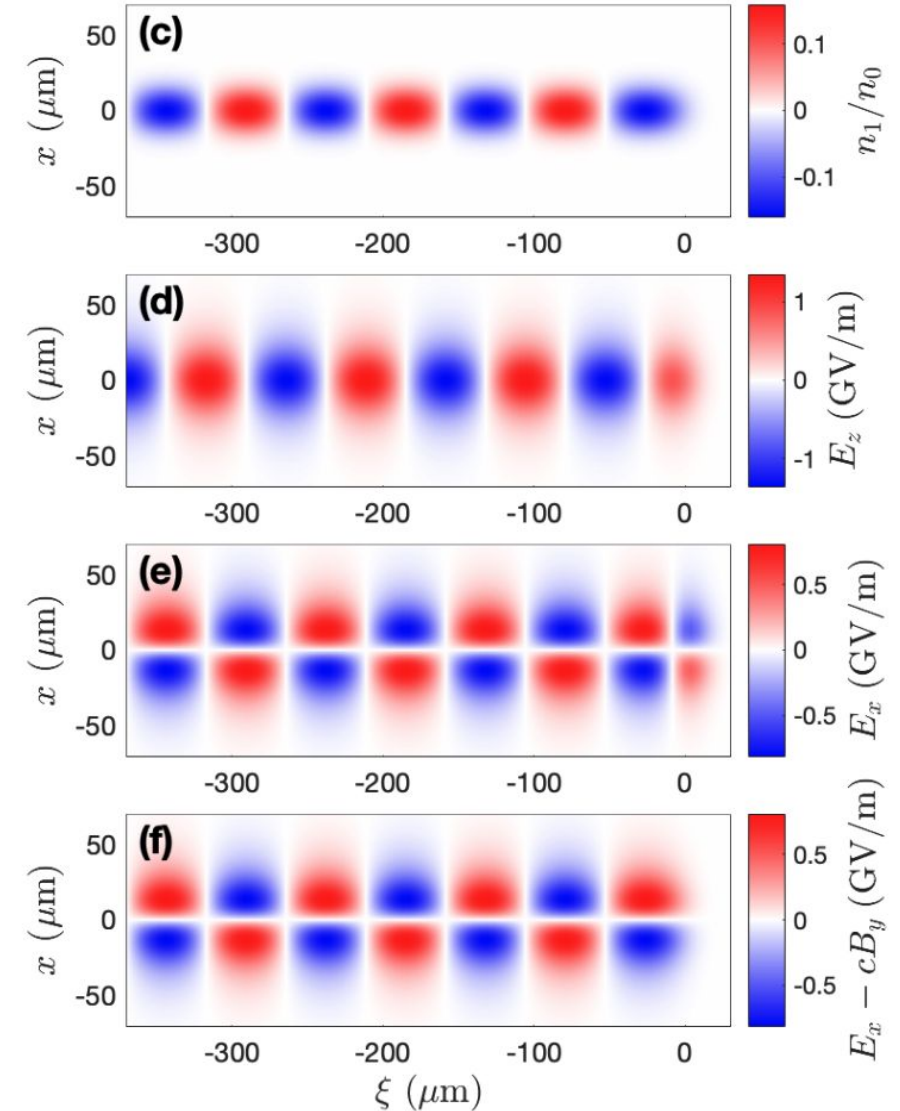
- Change to co-moving coordinates:

$$\frac{\partial^2 n_1}{\partial t^2} = -\omega_p^2 n_1 - \omega_p^2 n_b$$

$$\frac{\partial^2 n_1}{\partial \xi^2} = -\frac{\omega_p^2}{c^2} n_1 - \frac{\omega_p^2}{c^2} n_b$$

$$\rightarrow \left(\frac{\partial^2}{\partial \xi^2} + k_p^2 \right) n_1 = -k_p^2 n_b$$

where $k_p = \omega_p/c$

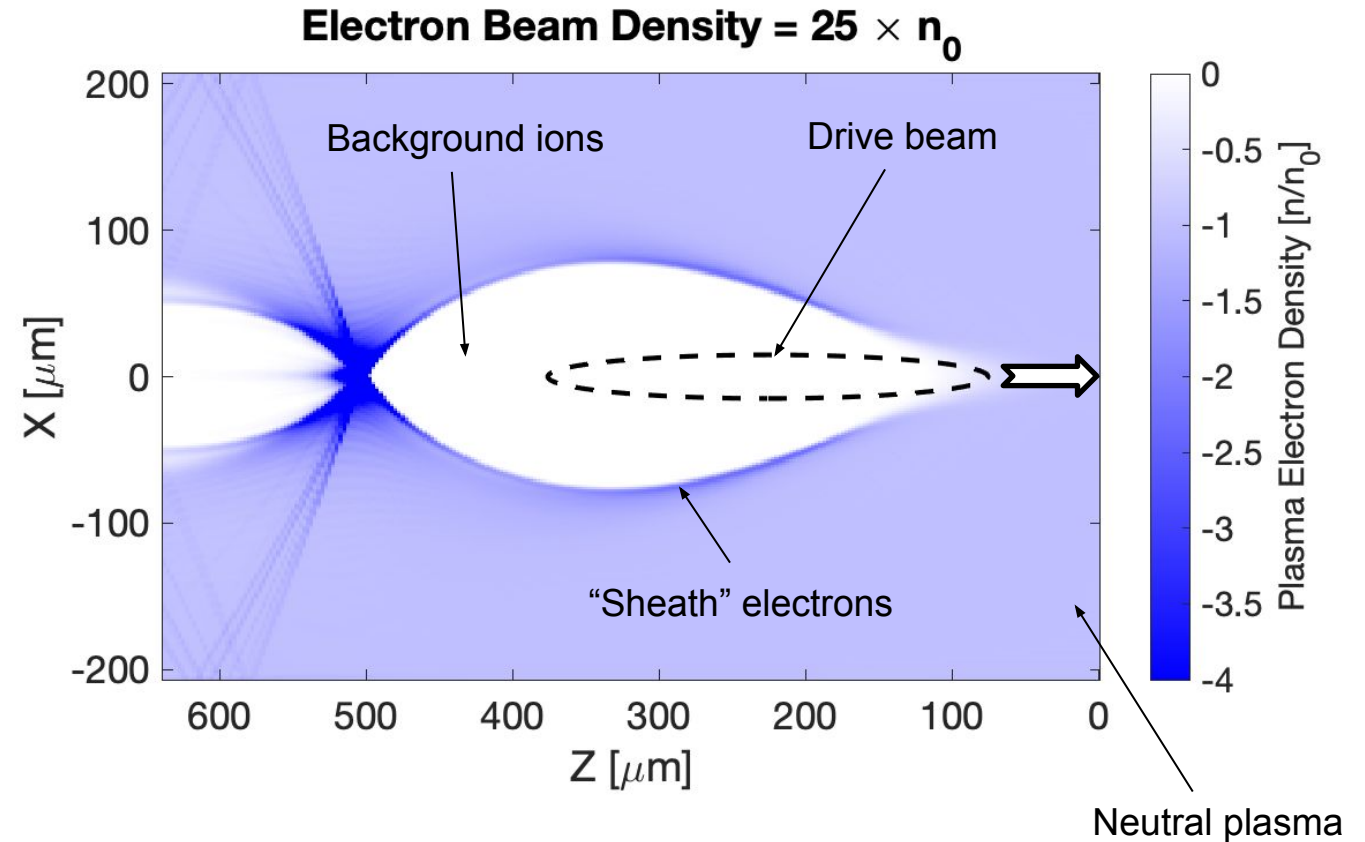


Limitations of the Linear Regime

To understand the limitations of the Linear Regime, let's quickly examine the nonlinear regime.

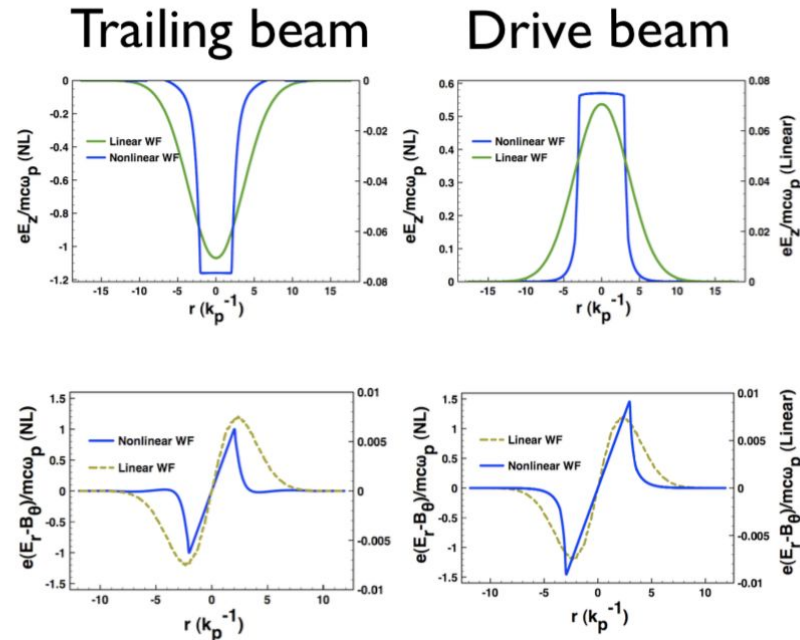
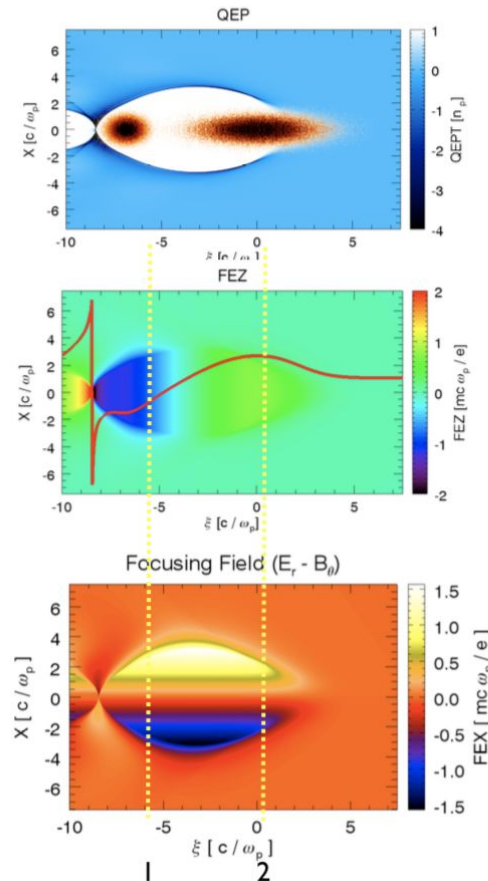
If the drive beam is much denser than the background plasma, the gently oscillating plasma wakefield turns into a plasma bubble.

We refer to this as the “**Bubble Regime**” or “**Blowout Regime**”.



Prof. Mori's 4 Conditions

Nonlinear wakefield is **IDEAL** for accelerating/focusing electrons
Trailing beam does not modify focusing fields of wake



$$\partial_\xi F_z = 0$$

$$\partial_\xi F_\perp = 0$$

$$\nabla_\perp F_\perp = C_{\text{constant}}$$

$$\nabla_\perp F_z = 0$$

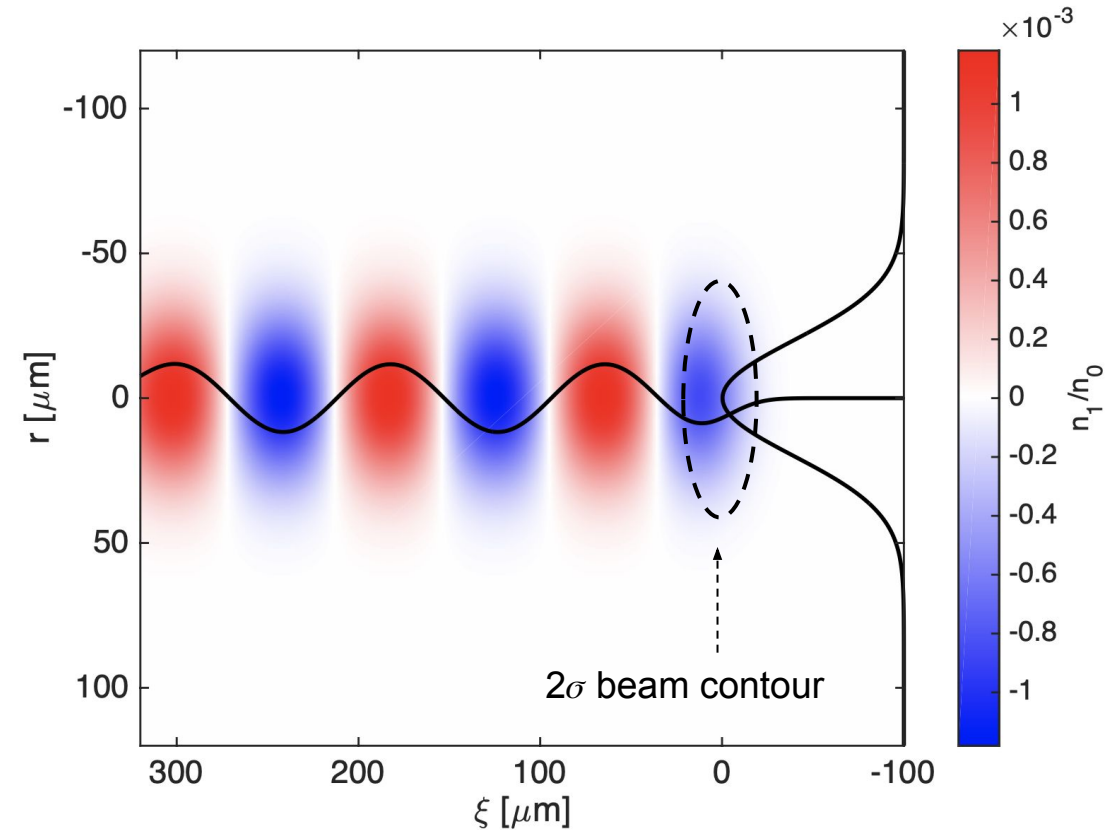
Back to the Linear Regime

The linear condition is: $\frac{n_b}{n_0} \ll 1$

The plasma density perturbation has the same transverse shape as the beam distribution:

$$n_1(r, \xi) = -k_p \boxed{R(r)} \int_{\xi}^{\infty} Z(\xi') \sin[k_p(\xi - \xi')] d\xi'$$

Transverse beam distribution



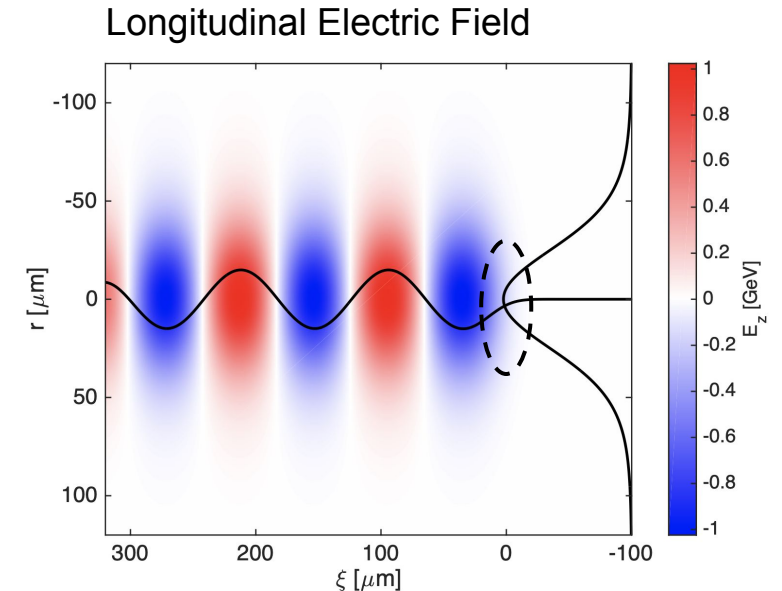
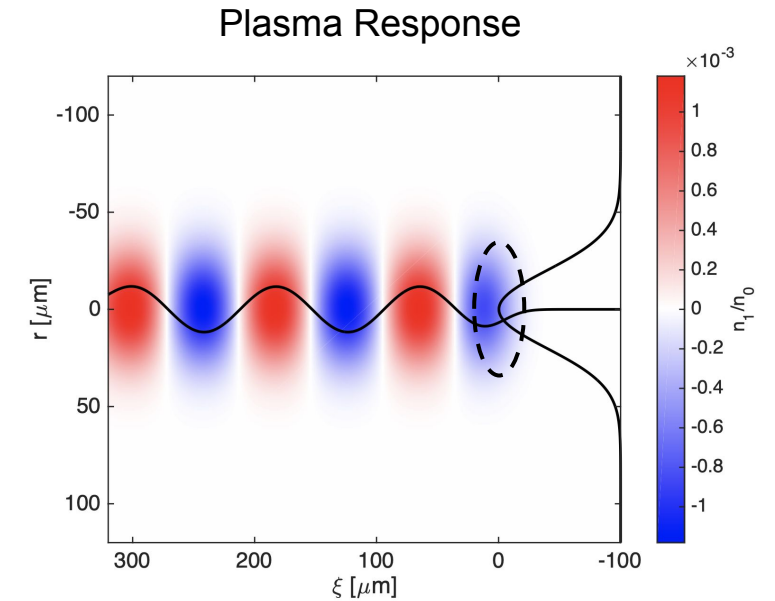
Fields due to Plasma Response

$$n_1(r, \xi) = -k_p \boxed{R(r)} \int_{\xi}^{\infty} Z(\xi') \sin [k_p(\xi - \xi')] d\xi'$$

Transverse beam distribution

$$E_z(r, \xi) = \frac{e}{4\pi\epsilon_0} k_p^2 \int_{\xi}^{\infty} \int_0^{\infty} \boxed{R(r') g_0(r, r_0)} Z(\xi') \cos [k_p(\xi - \xi')] r' dr' d\xi'$$

Convolution of transverse beam distribution with Bessel functions of order 0



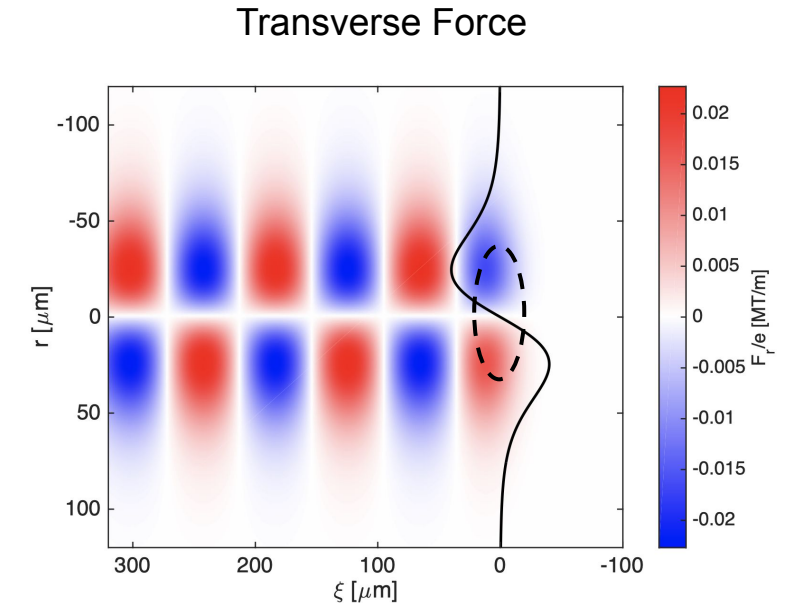
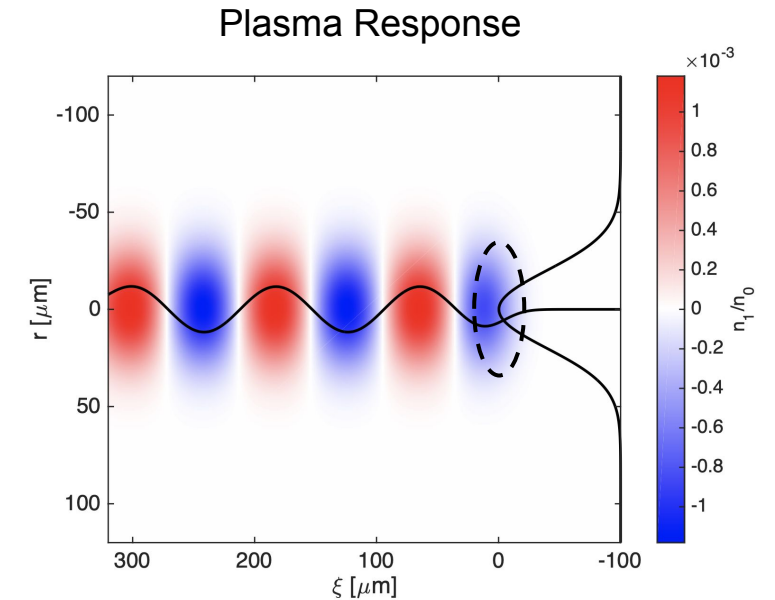
Fields due to Plasma Response

$$n_1(r, \xi) = -k_p \boxed{R(r)} \int_{\xi}^{\infty} Z(\xi') \sin [k_p(\xi - \xi')] d\xi'$$

Transverse beam distribution

$$F_r(r, \xi) = -ek_p \int_{\xi}^{\infty} \int_0^{\infty} \boxed{\frac{\partial R(r')}{\partial r} g_1(r, r')} Z(\xi') \sin k_p(\xi - \xi') r' dr' d\xi'$$

Convolution of derivative of
transverse beam distribution with
Bessel functions of order 1



What defines a “good” particle accelerator?

A good particle accelerator...

- Provides the same accelerating field to different particles in the bunch, regardless of where they are in the bunch.
- Maintains the emittance of the accelerated bunch.

The 4 Conditions

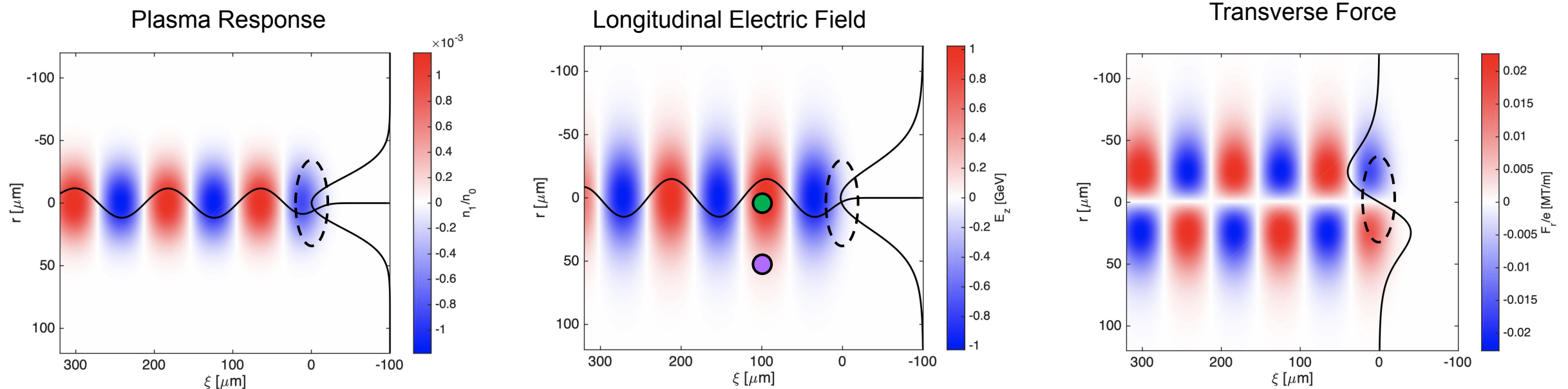
$$(1) \quad \partial_{\xi} F_z = 0$$

$$(2) \quad \partial_{\xi} F_{\perp} = 0$$

$$(3) \quad \nabla_{\perp} F_{\perp} = C$$

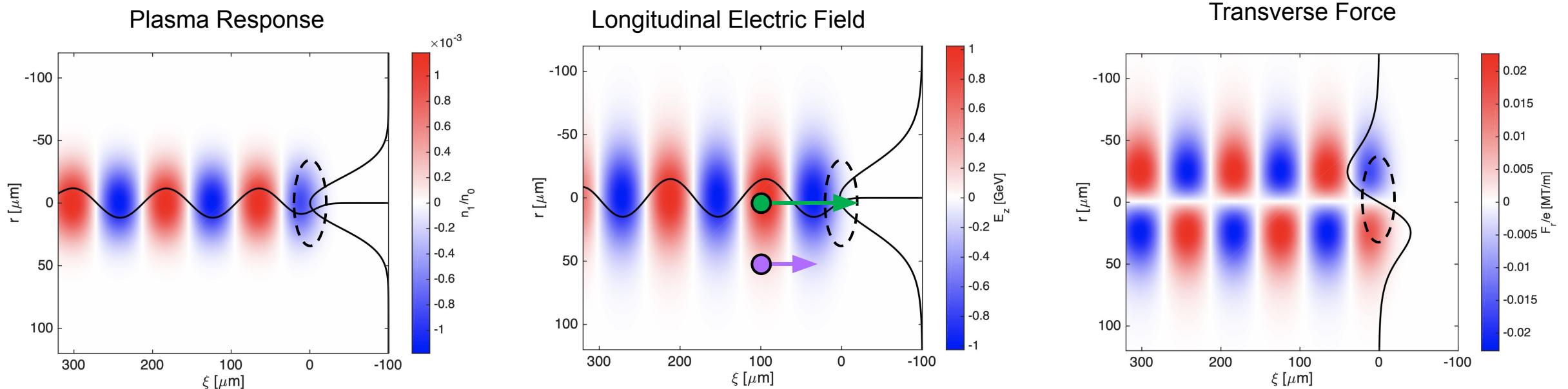
$$(4) \quad \nabla_{\perp} F_z = 0$$

Limitations of Linear Plasma Wakefields



Consider **on-axis** and **off-axis** test particles.

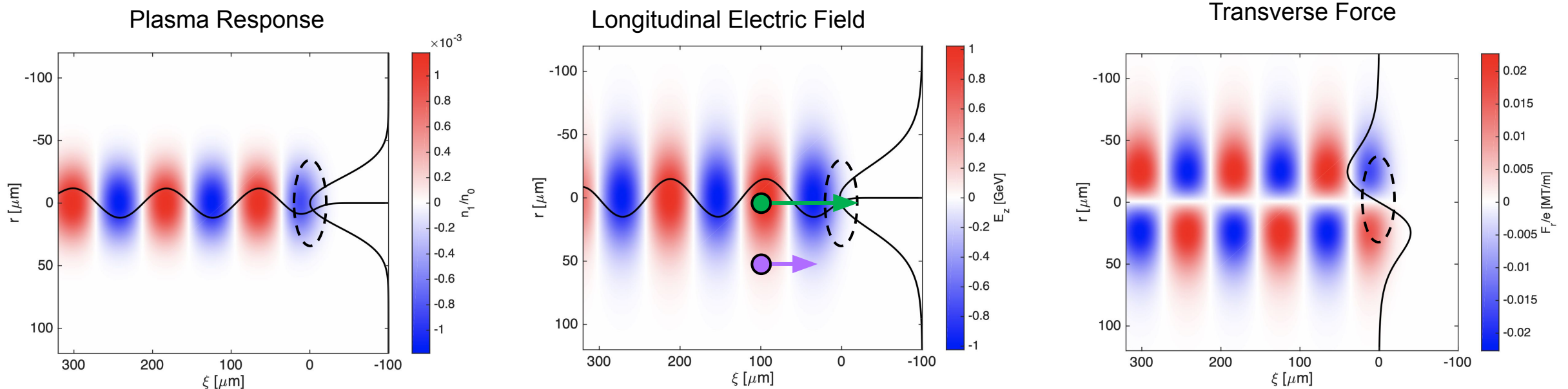
Limitations of Linear Plasma Wakefields



Consider **on-axis** and **off-axis** test particles.

The **on-axis** particle experiences a larger accelerating field than the **off-axis** particle.

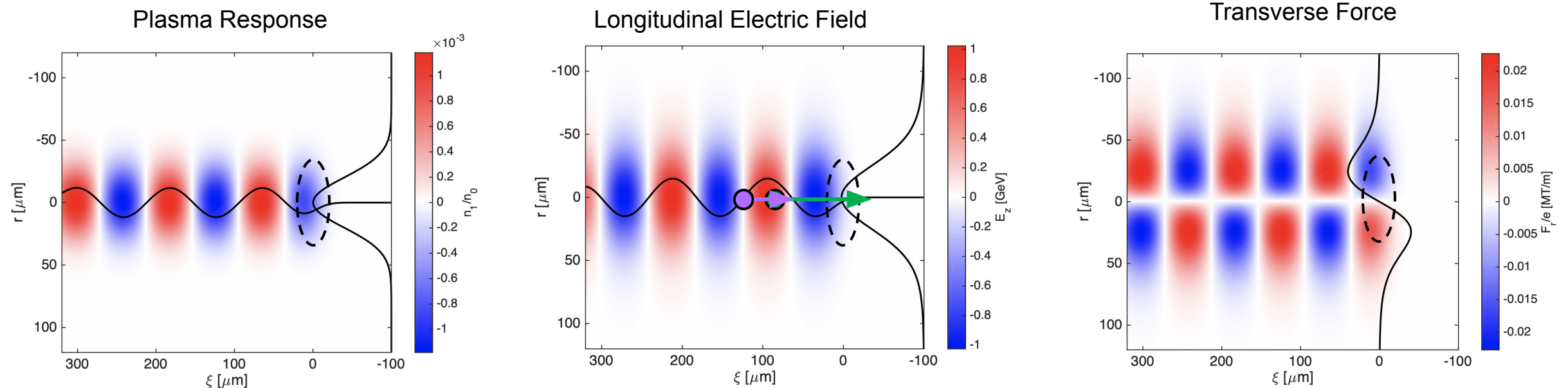
Limitations of Linear Plasma Wakefields



Consider **on-axis** and **on-axis** test particles.

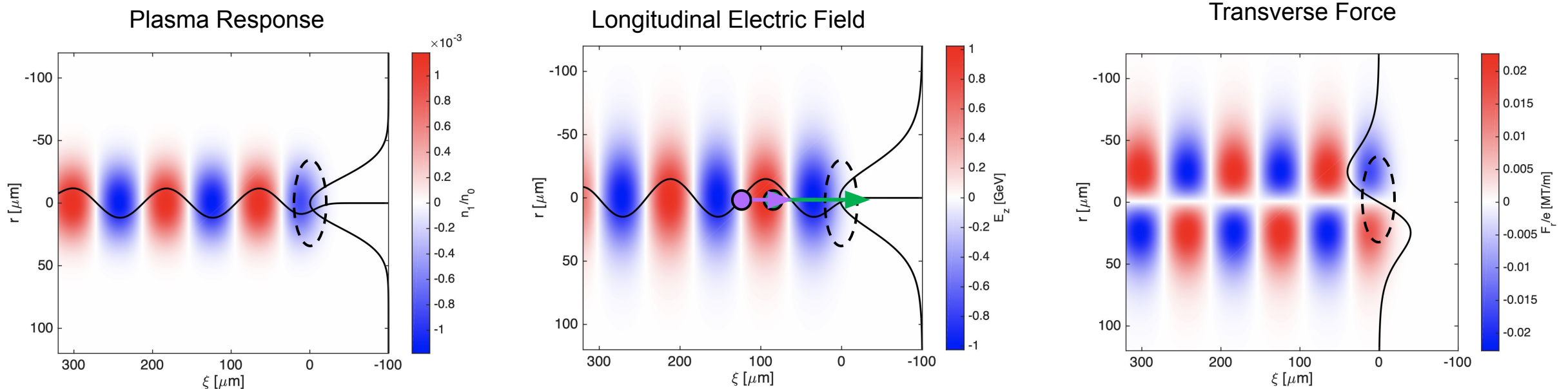
Condition (4) $\nabla_{\perp} F_z = 0$ is not satisfied!

Limitations of Linear Plasma Wakefields



Consider **on-axis-ahead** and **on-axis-behind** test particles.

Limitations of Linear Plasma Wakefields

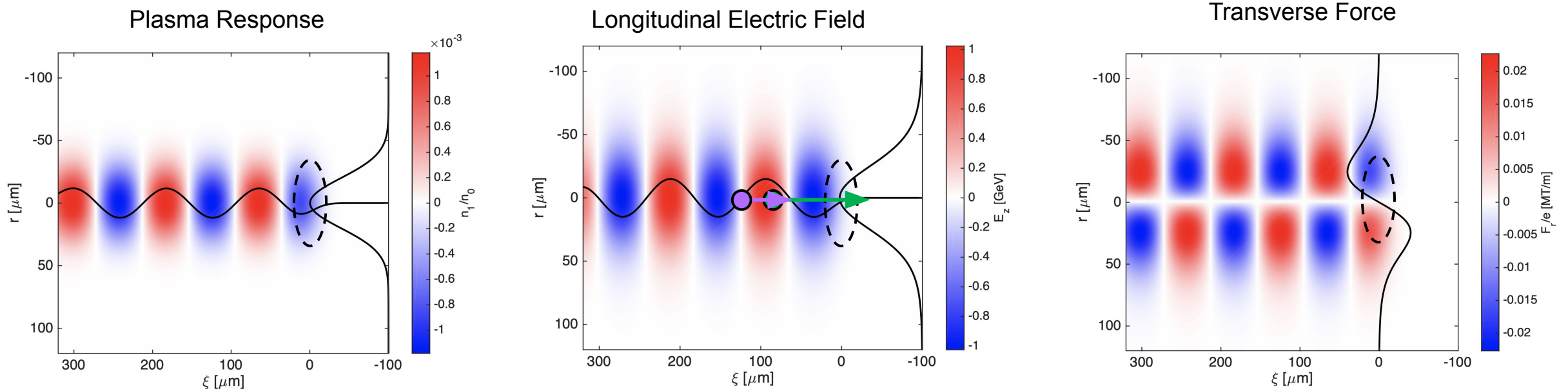


Consider **on-axis-ahead** and **on-axis-behind** test particles.

The **on-axis-ahead** particle experiences a larger accelerating field than the **on-axis-behind** particle*.

*We are not considering **beam loading** effects.

Limitations of Linear Plasma Wakefields

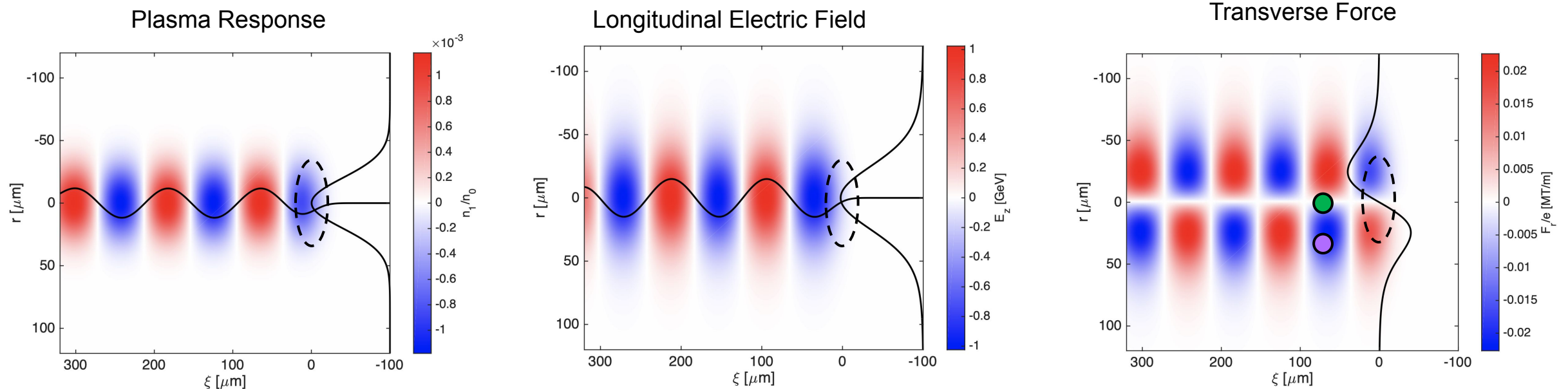


Consider **on-axis-ahead** and **on-axis-behind** test particles.

Condition (1) $\partial_{\xi} F_z = 0$ not satisfied*

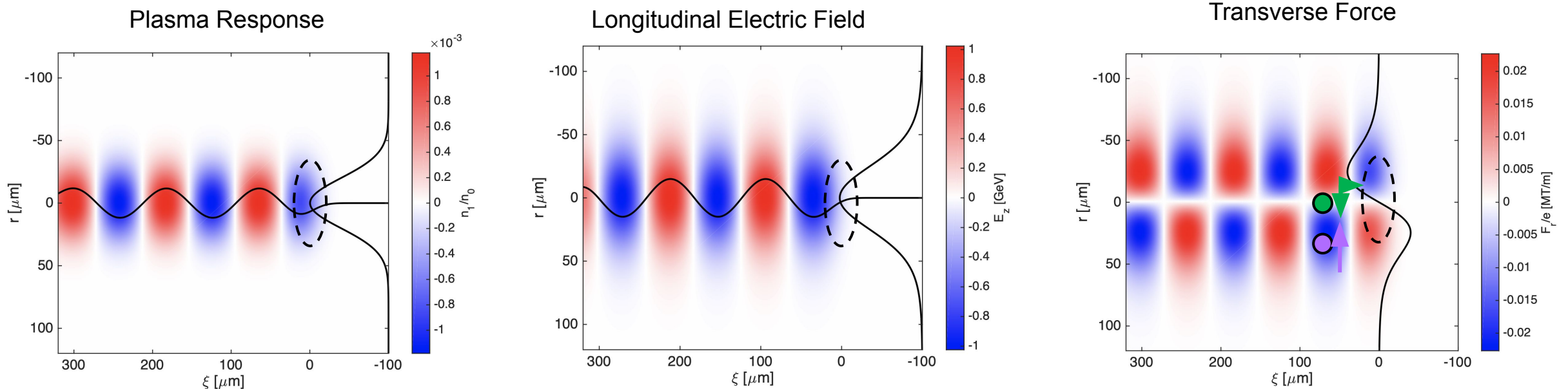
*We are not considering **beam loading** effects.

Limitations of Linear Plasma Wakefields



Consider **on-axis** and **off-axis** test particles.

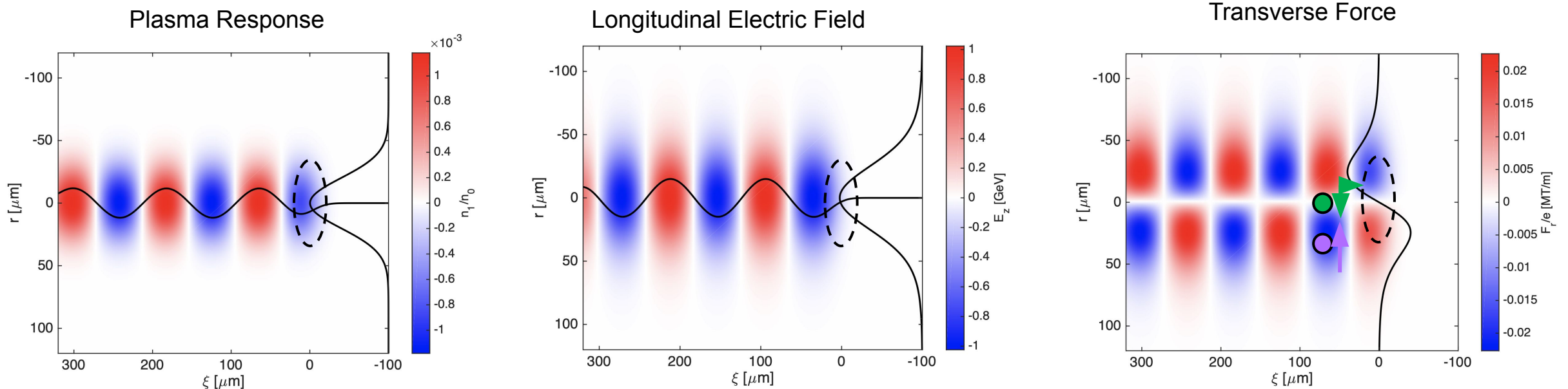
Limitations of Linear Plasma Wakefields



Consider **on-axis** and **off-axis** test particles.

The **on-axis** particle experiences a linear focusing force, while the **off-axis** particle experiences amplitude-dependant oscillations.

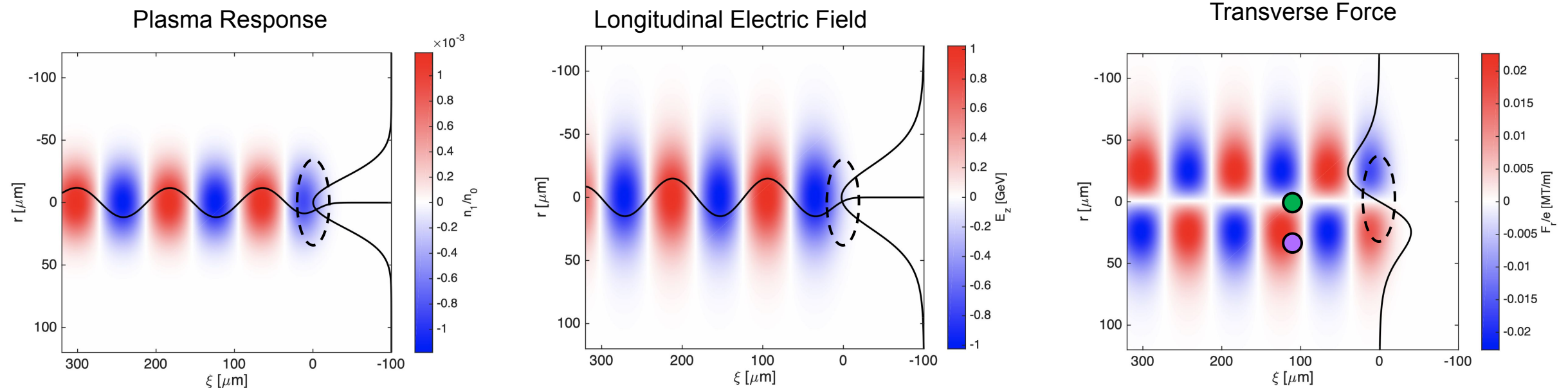
Limitations of Linear Plasma Wakefields



Consider **on-axis** and **off-axis** test particles.

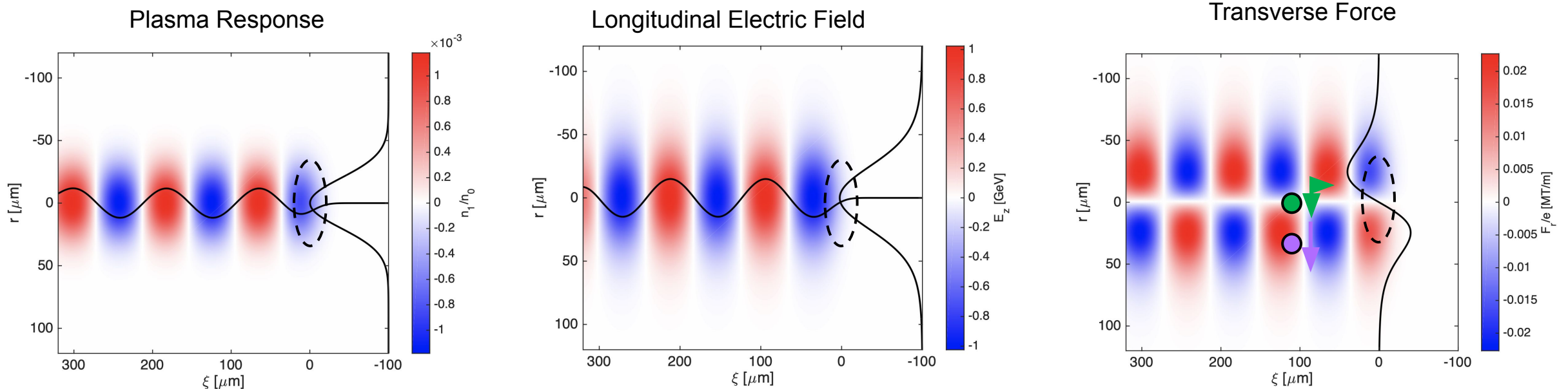
Condition (3) $\nabla_{\perp} F_{\perp} = C$ not satisfied!

Limitations of Linear Plasma Wakefields



Consider **on-axis** and **off-axis** test particles at a different position in the wake.

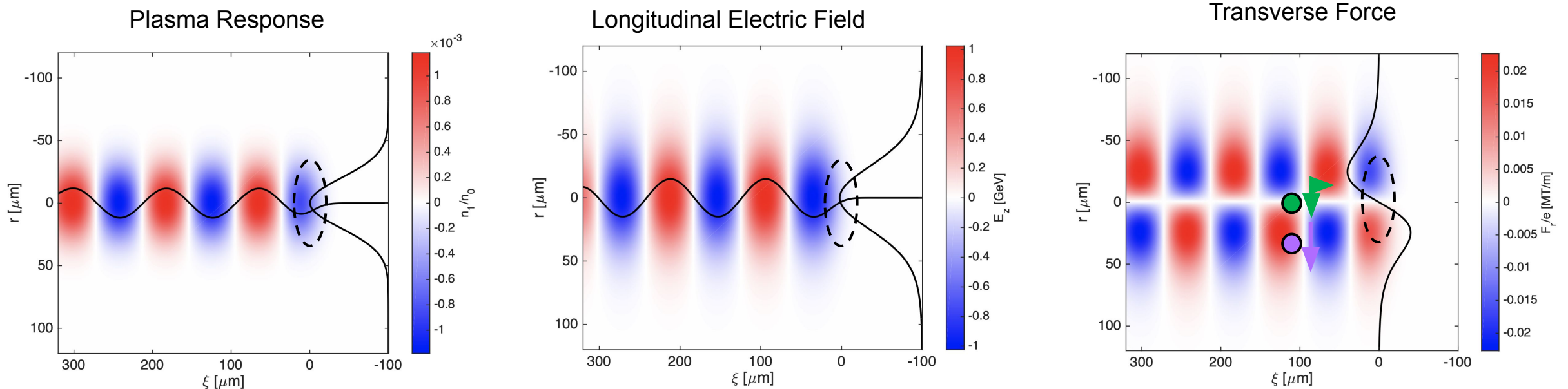
Limitations of Linear Plasma Wakefields



Consider **on-axis** and **off-axis** test particles at a different position in the wake.

Here, the transverse force is *defocusing*!

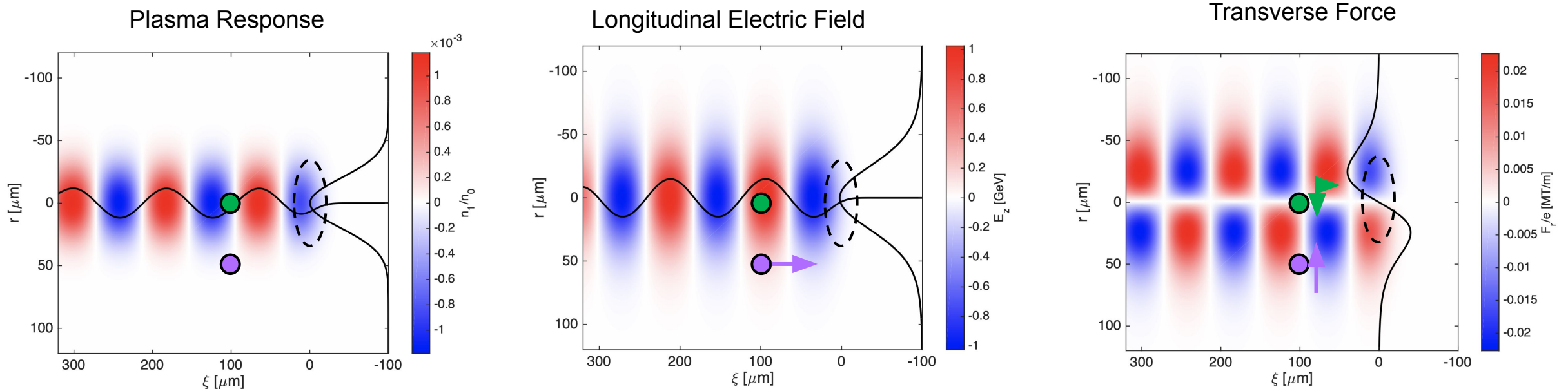
Limitations of Linear Plasma Wakefields



Consider **on-axis** and **off-axis** test particles at a different position in the wake.

Condition (2) $\partial_{\xi} F_{\perp} = 0$ not satisfied!

Limitations of Linear Plasma Wakefields



In general, linear plasma wakefields do not provide uniform acceleration and focusing to particles in the wake! They are not ideal accelerators!

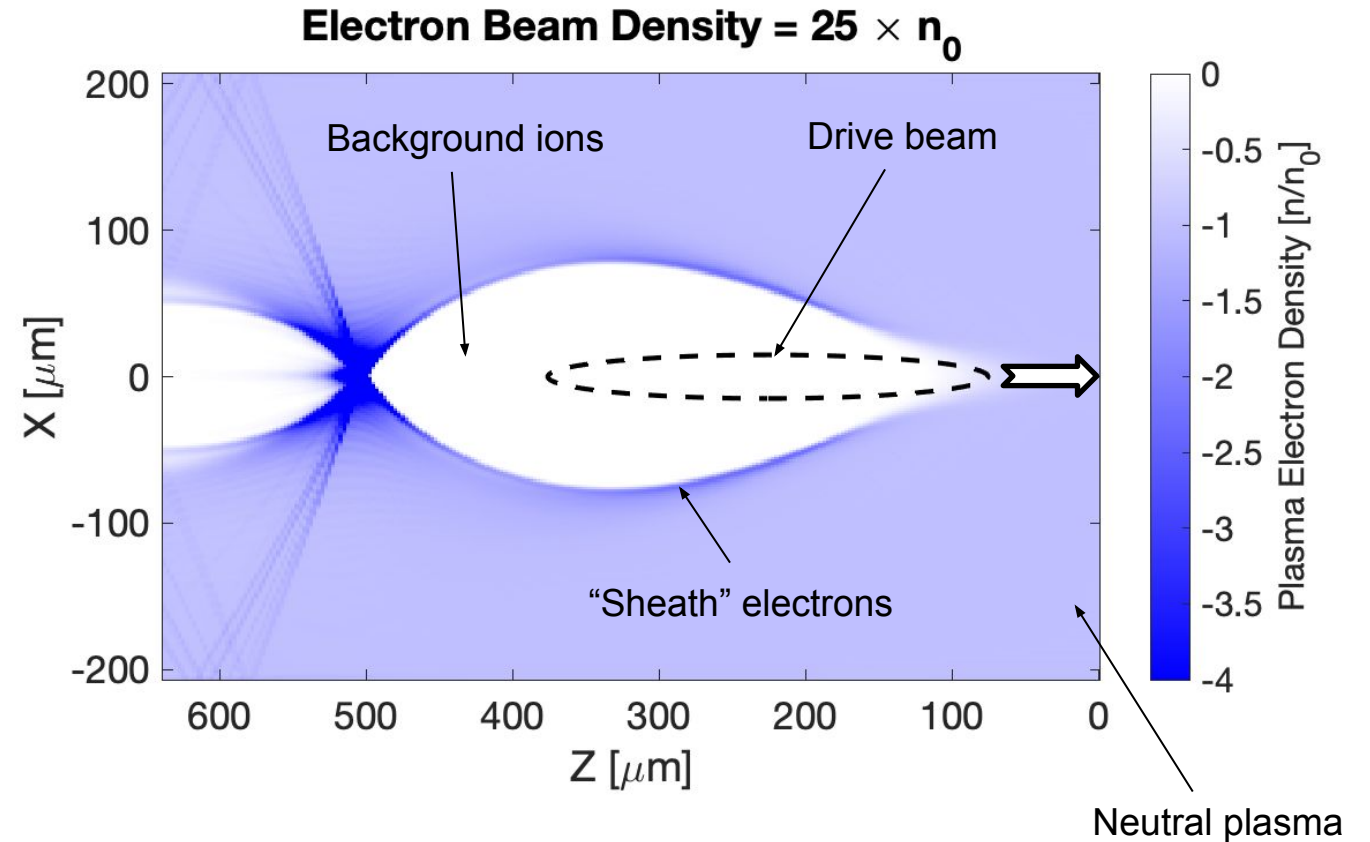
However, we can change the shape of the wake through the beam loading phenomenon. We will return to this point in the next lecture.



Nonlinear Regime

Fields in the nonlinear regime

How does the nonlinear regime provide “ideal” accelerating and focusing forces?



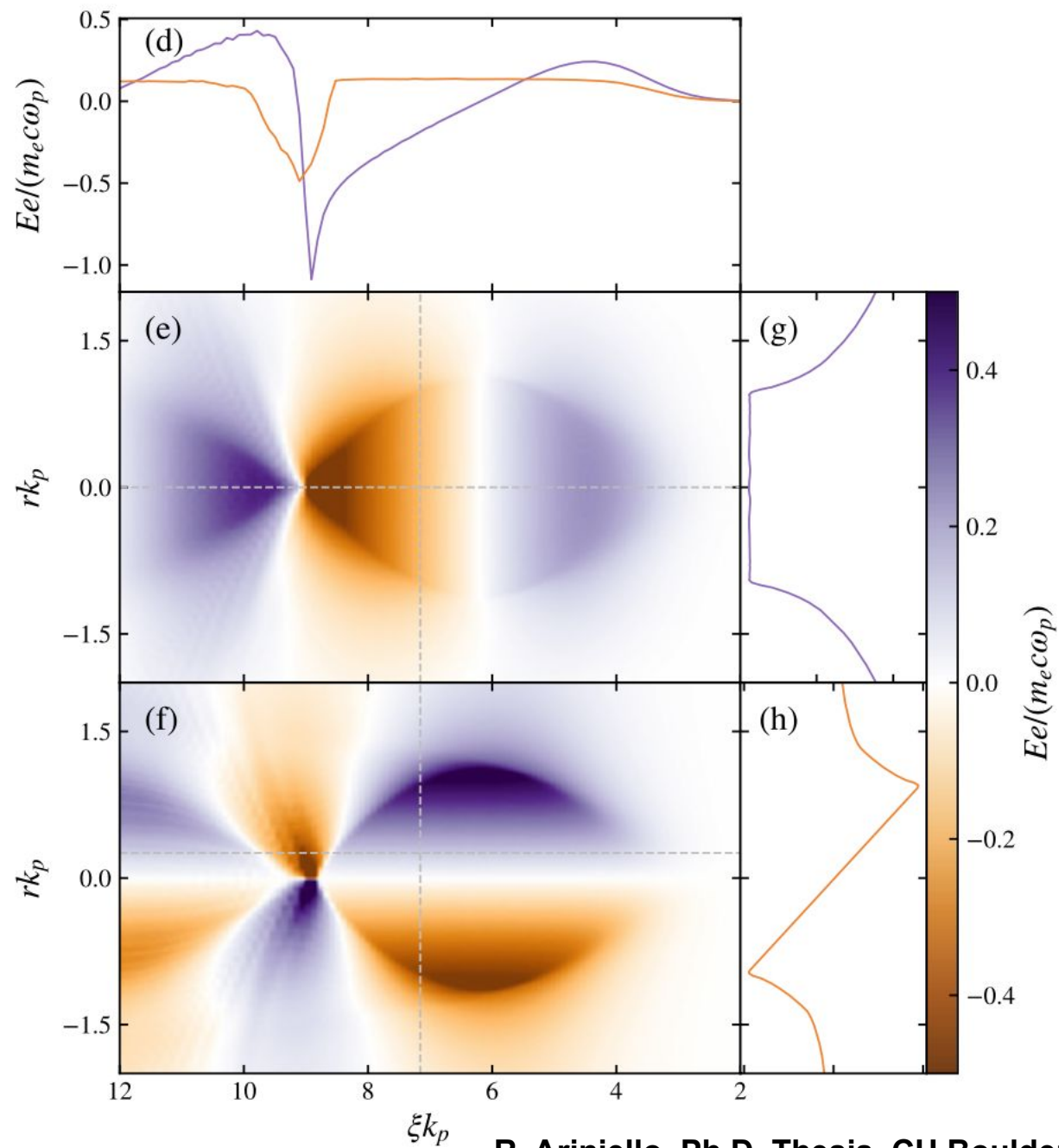
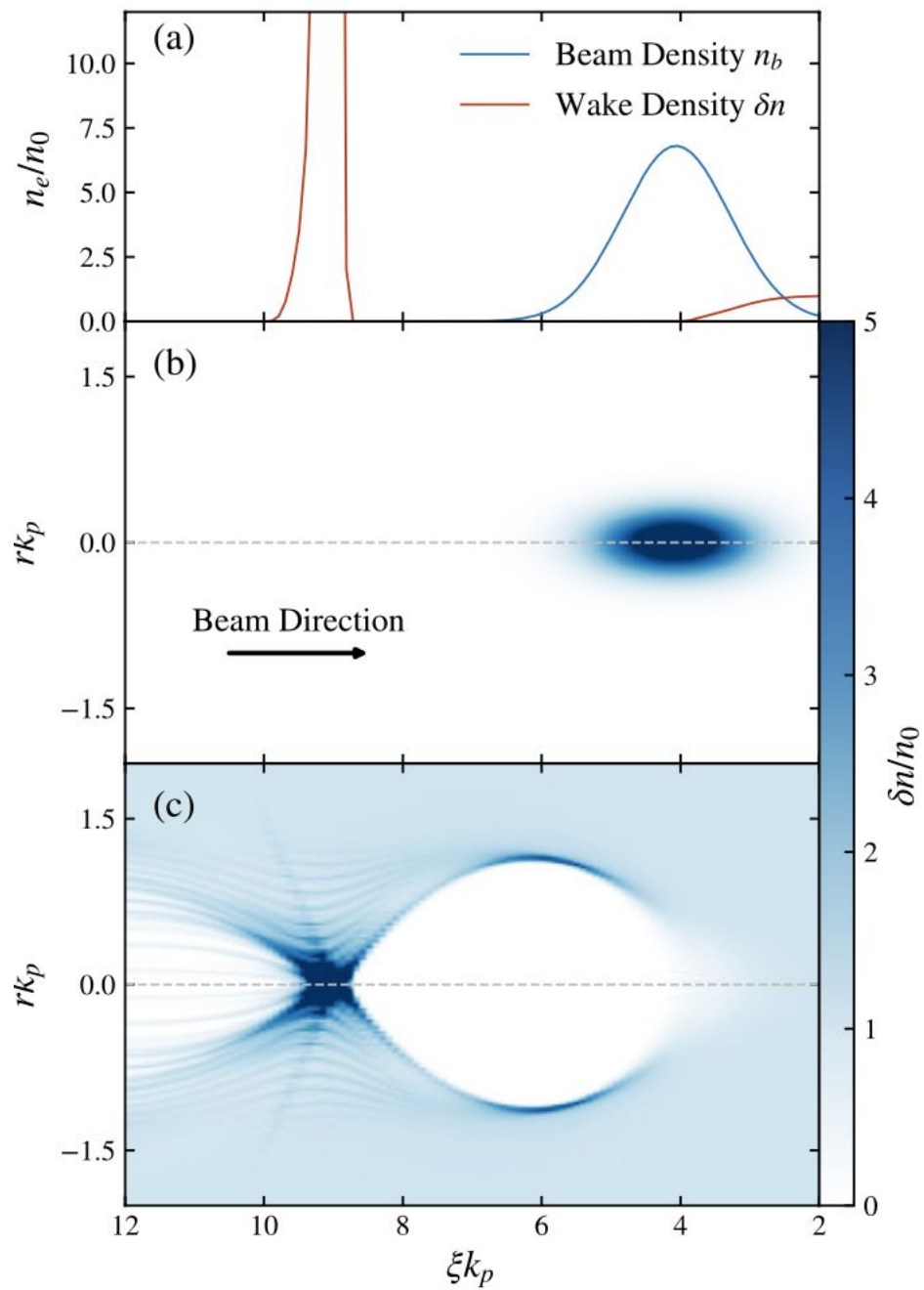
The 4 Conditions

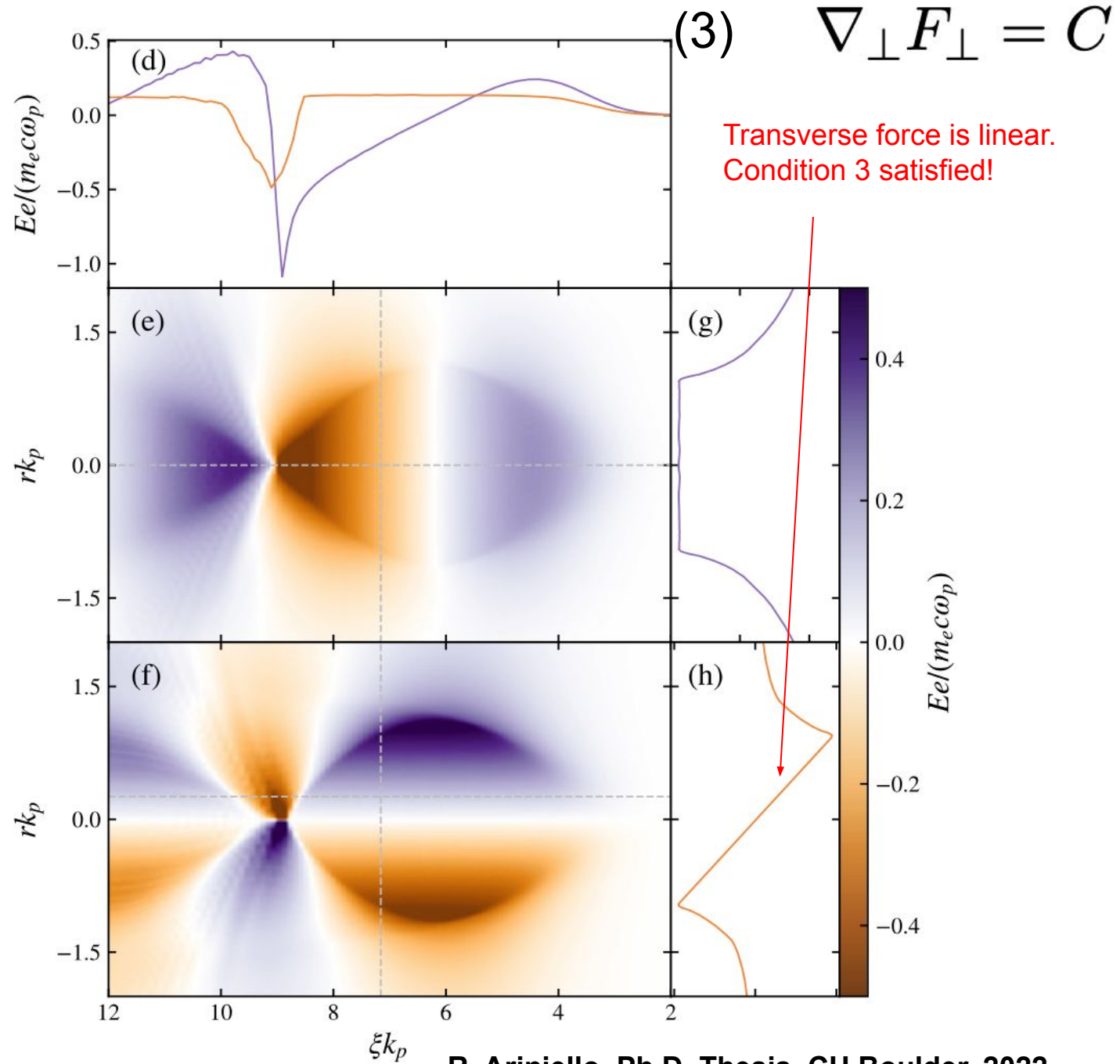
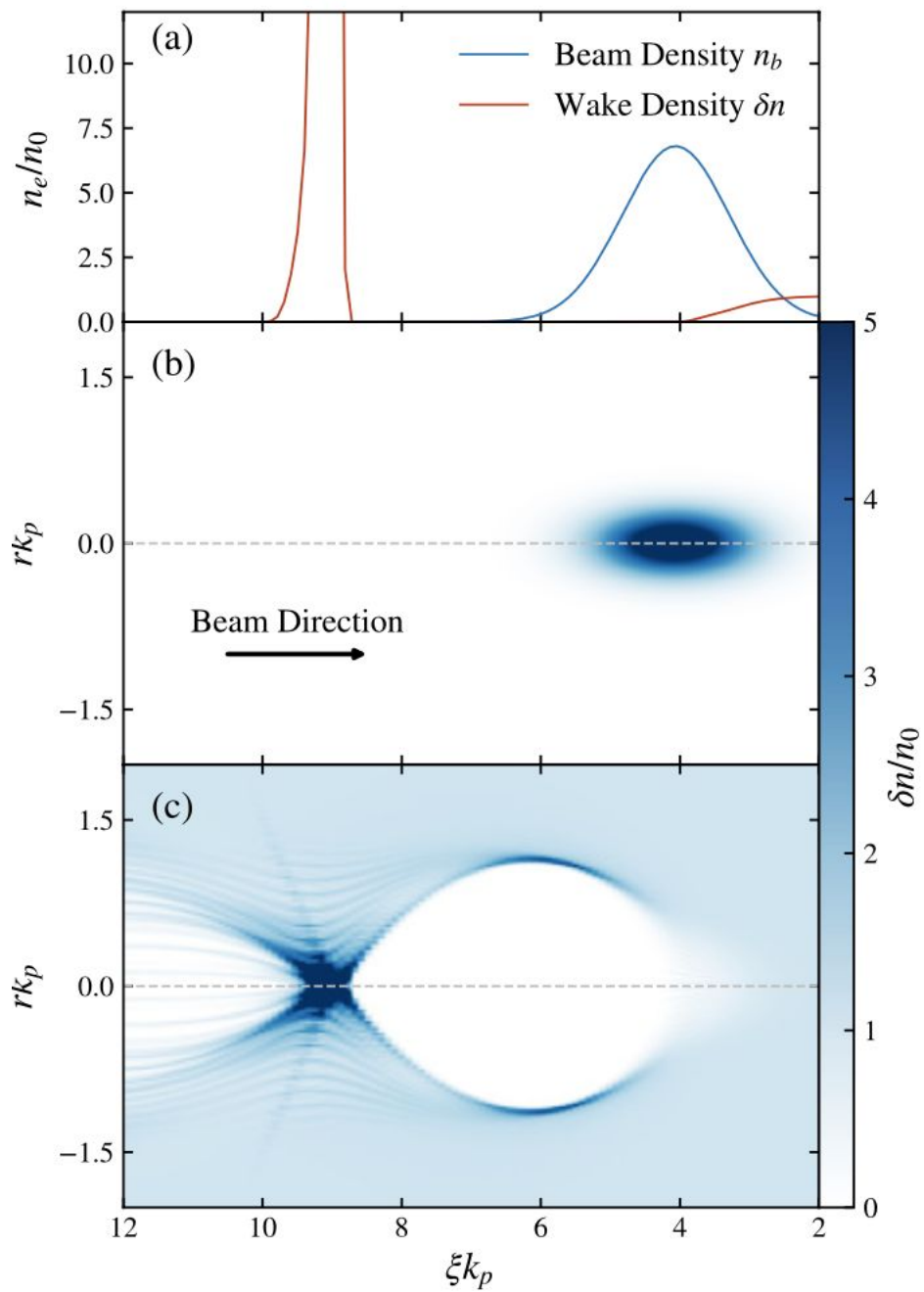
$$(1) \quad \partial_{\xi} F_z = 0$$

$$(2) \quad \partial_{\xi} F_{\perp} = 0$$

$$(3) \quad \nabla_{\perp} F_{\perp} = C$$

$$(4) \quad \nabla_{\perp} F_z = 0$$





Linear Transverse Focusing

Coulomb Field from Ions 2

- Gauss's Law

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

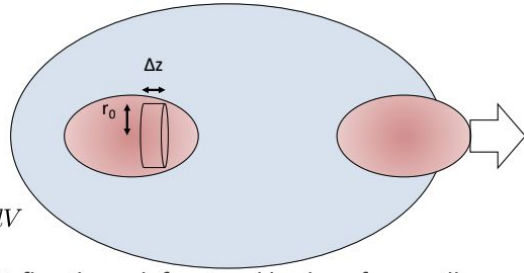
- Integral form:

$$\oint \vec{E} \cdot d\vec{S} = \int \frac{\rho}{\epsilon_0} dV$$

- Assume $\Delta z \rightarrow 0$: E_z flux through front and back surfaces will cancel each other

- Only term on LHS of eqn. is from radial E_r flux from ions

$$E_r 2\pi r_0 \Delta z = \frac{en_0}{\epsilon_0} \pi r_0^2 \Delta z$$



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Coulomb Field from Ions 3

- Solve for E_r at r_0 : $E_r(r_0) = \frac{en_0}{2\epsilon_0} r_0$

- Radial Coulomb force from ions at arbitrary position $r < R_{\text{wake}}$:

$$E_r(r) = \frac{en_0}{2\epsilon_0} r$$

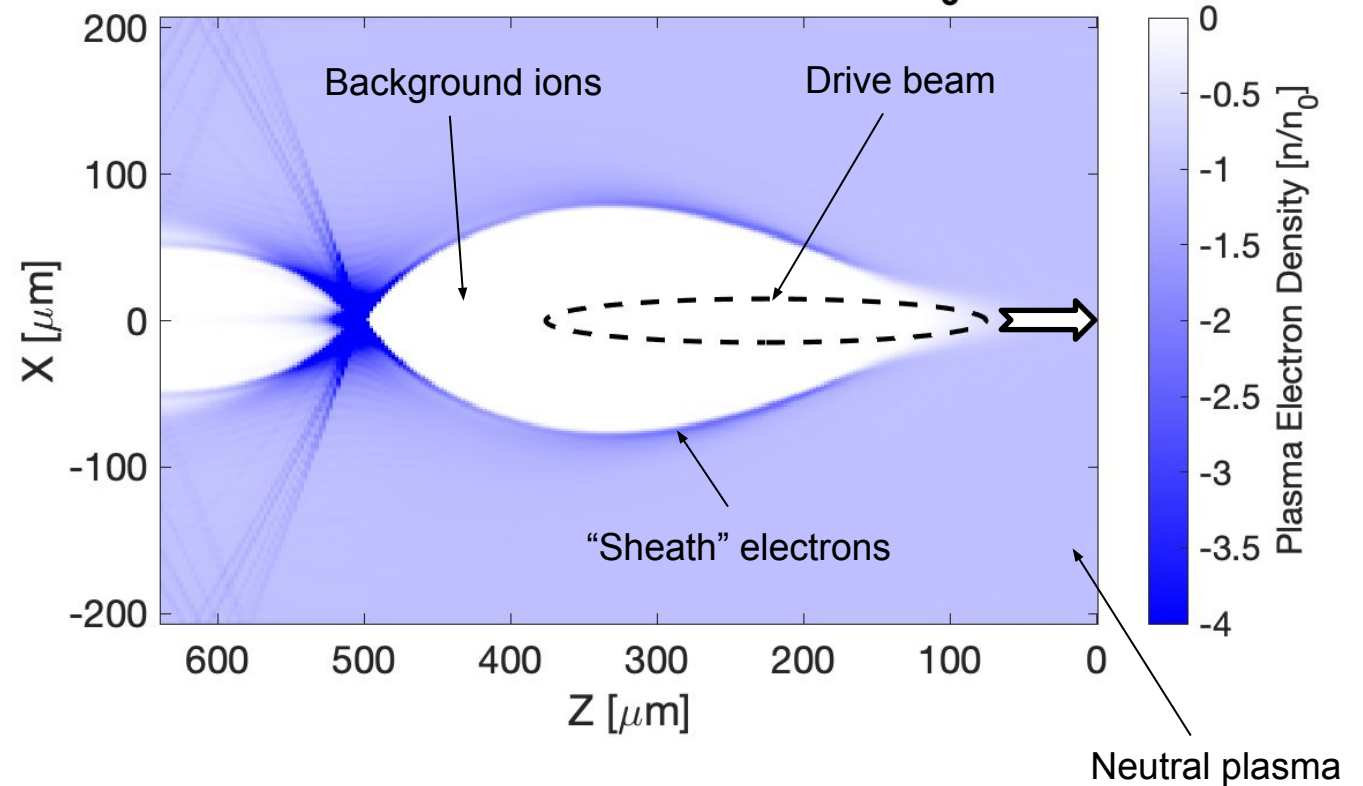
- Transverse force on a single beam electron:

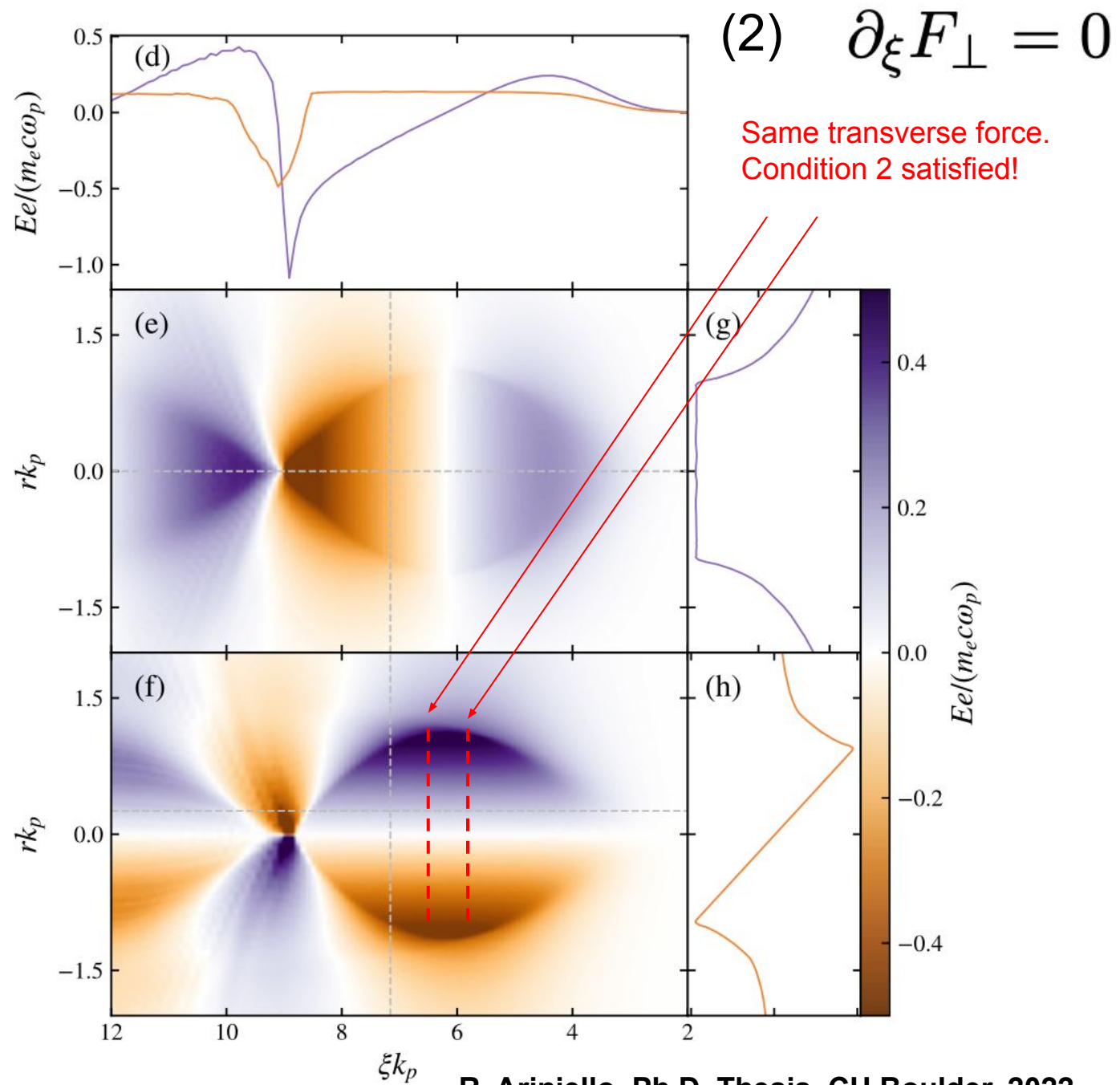
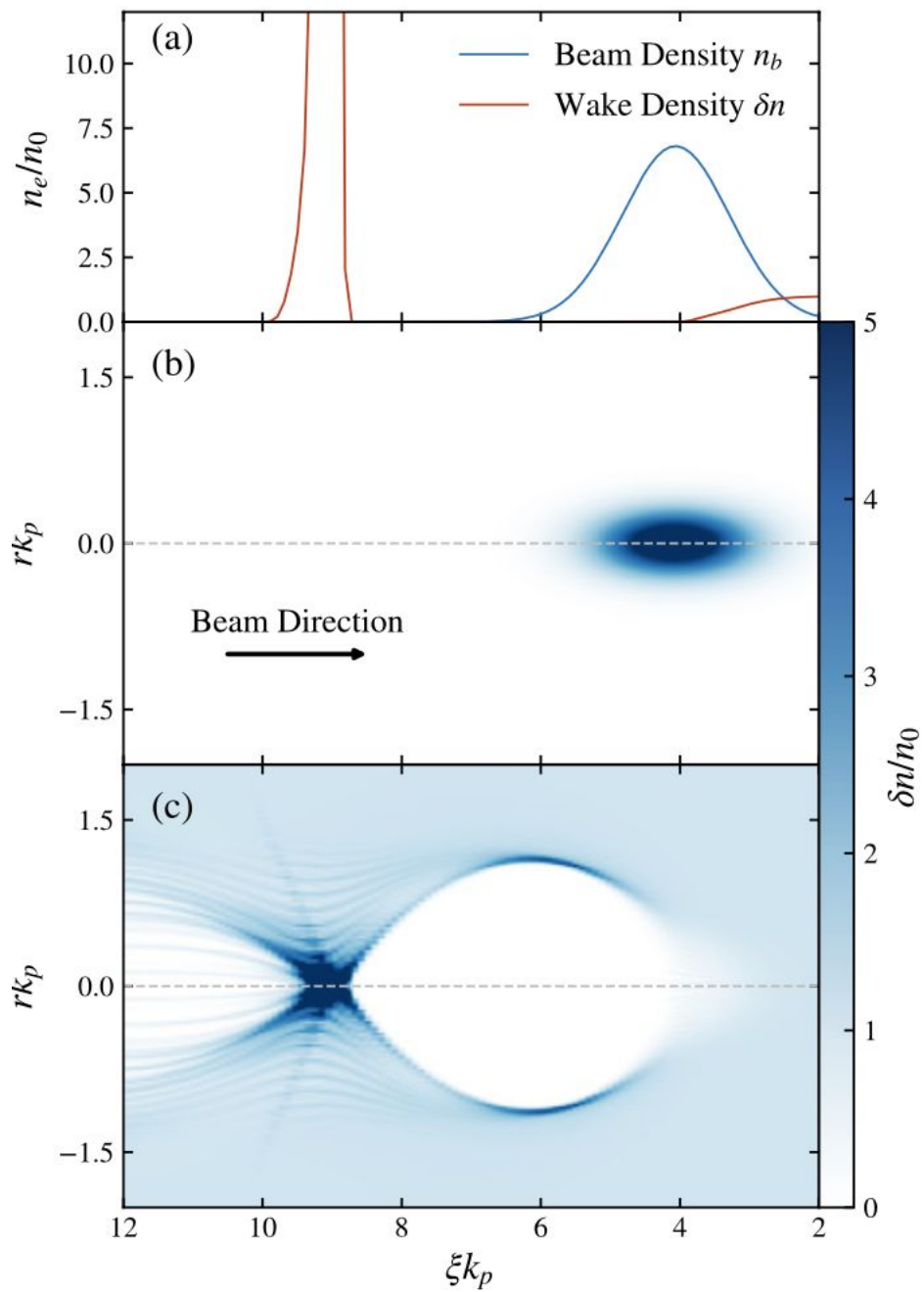
$$F_r(r) = \frac{dp_r}{dt} = \frac{d}{dt} \left(\gamma_b m_e \frac{dr}{dt} \right) = \gamma_b m_e \frac{d^2 r}{dt^2} = -e E_r(r)$$

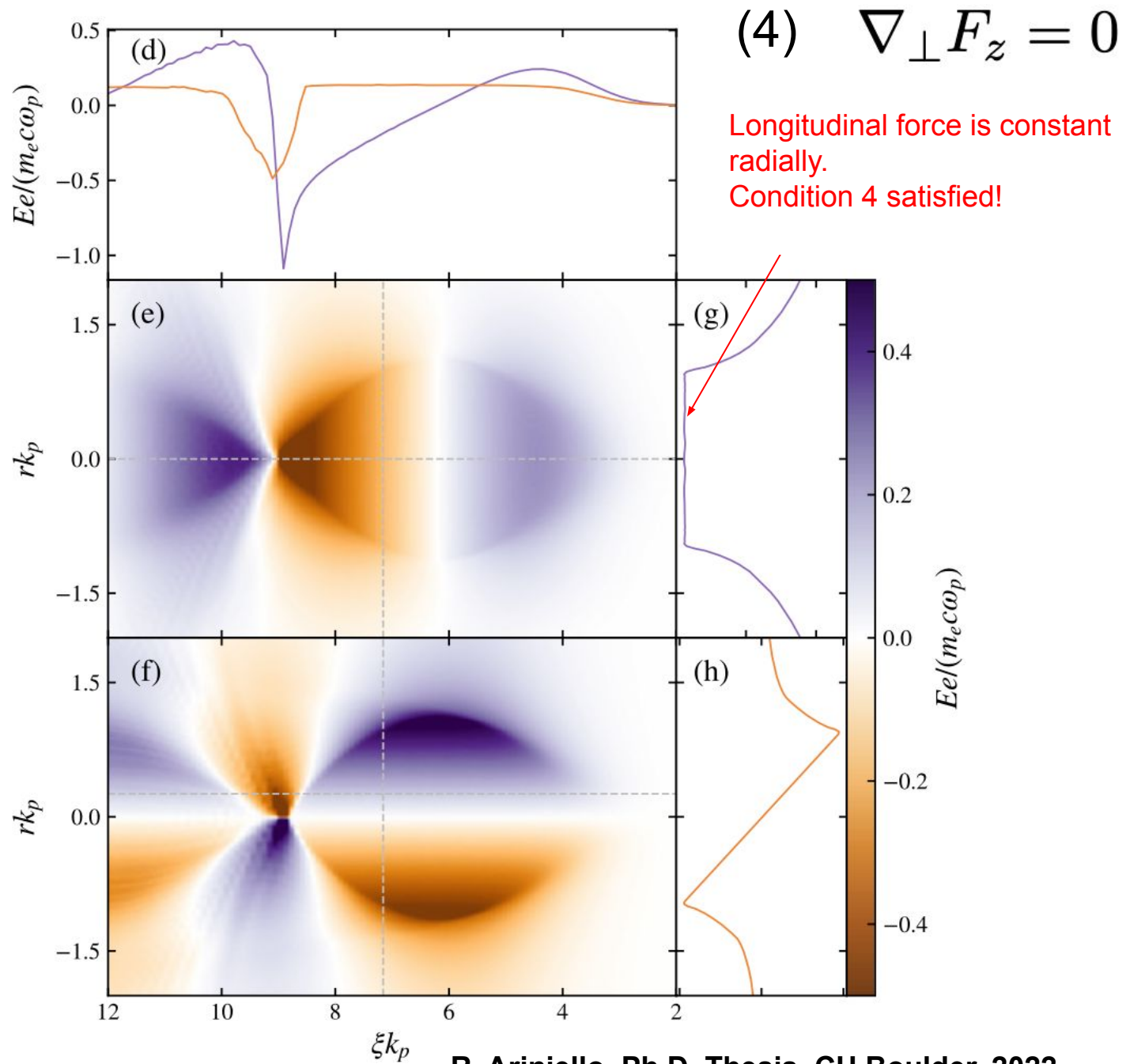
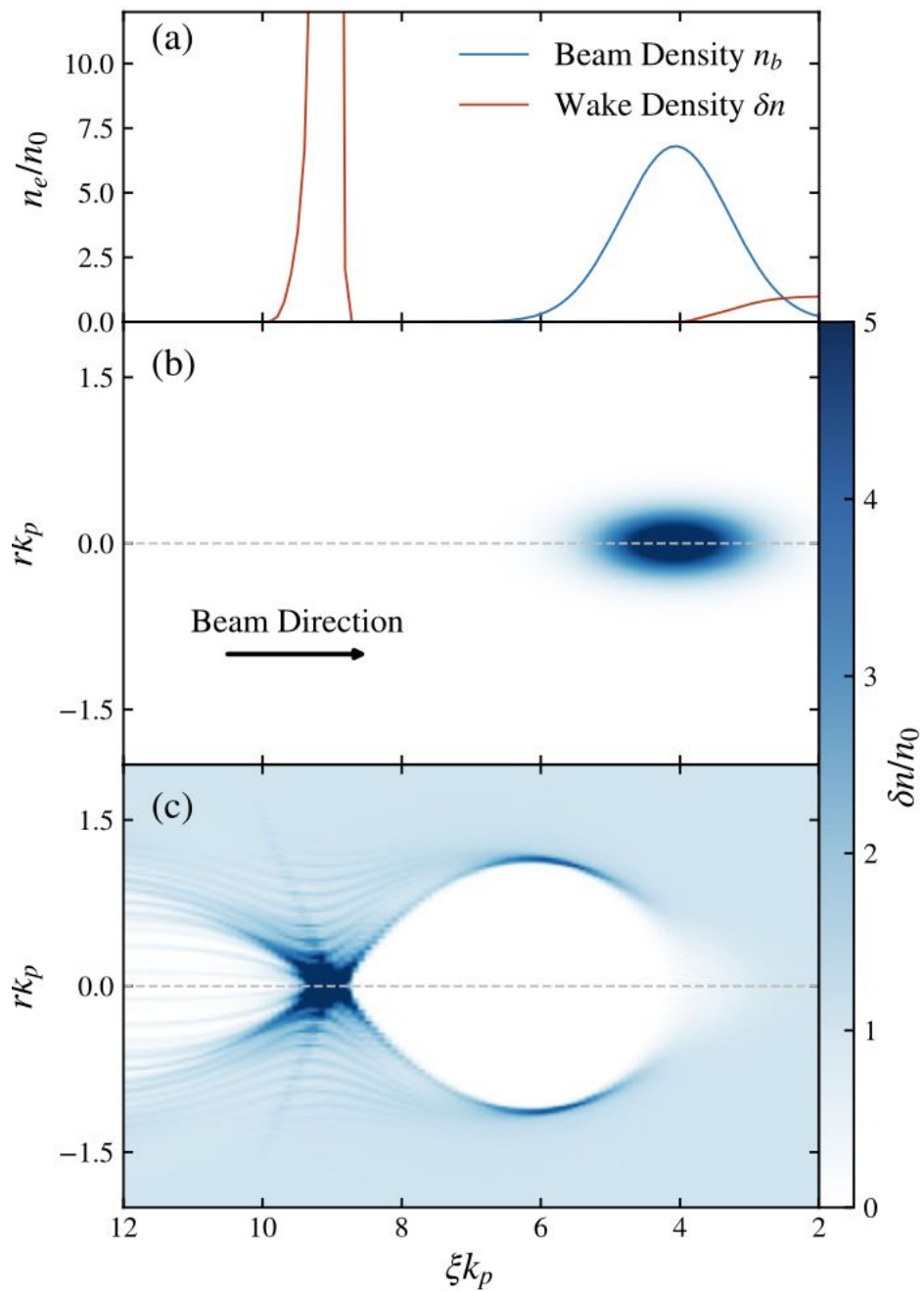
Lorentz factor

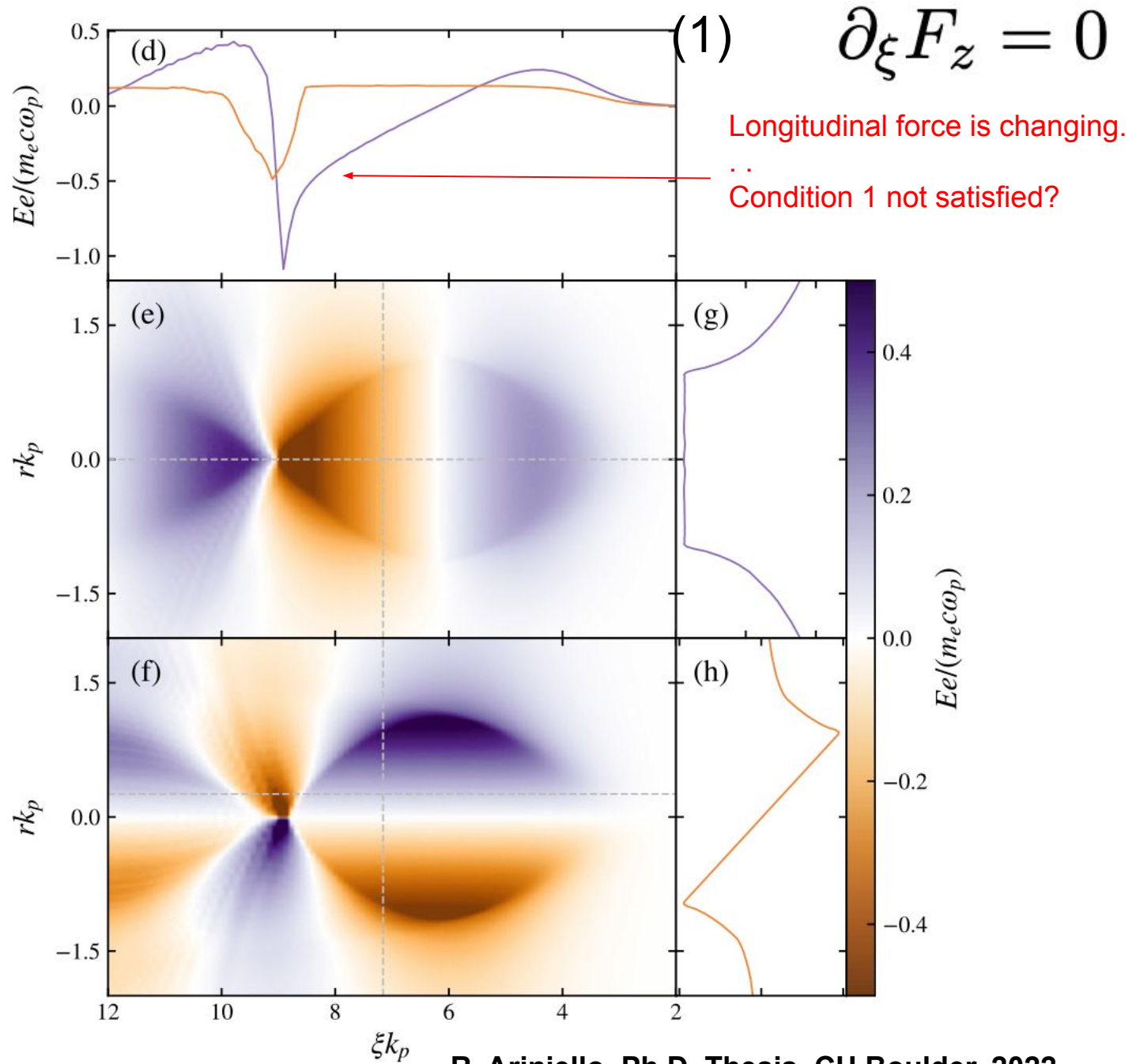
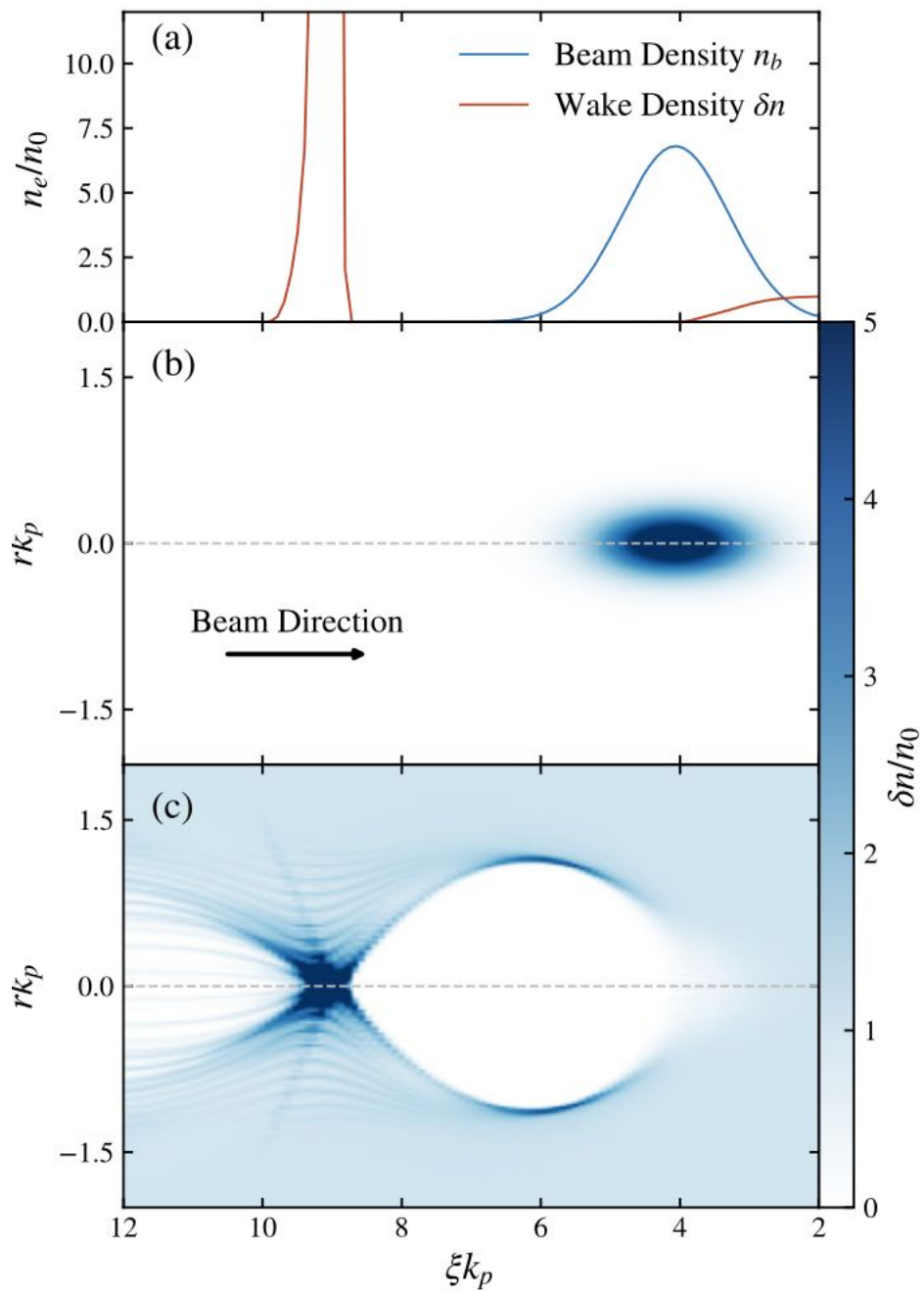
27

Electron Beam Density = $25 \times n_0$





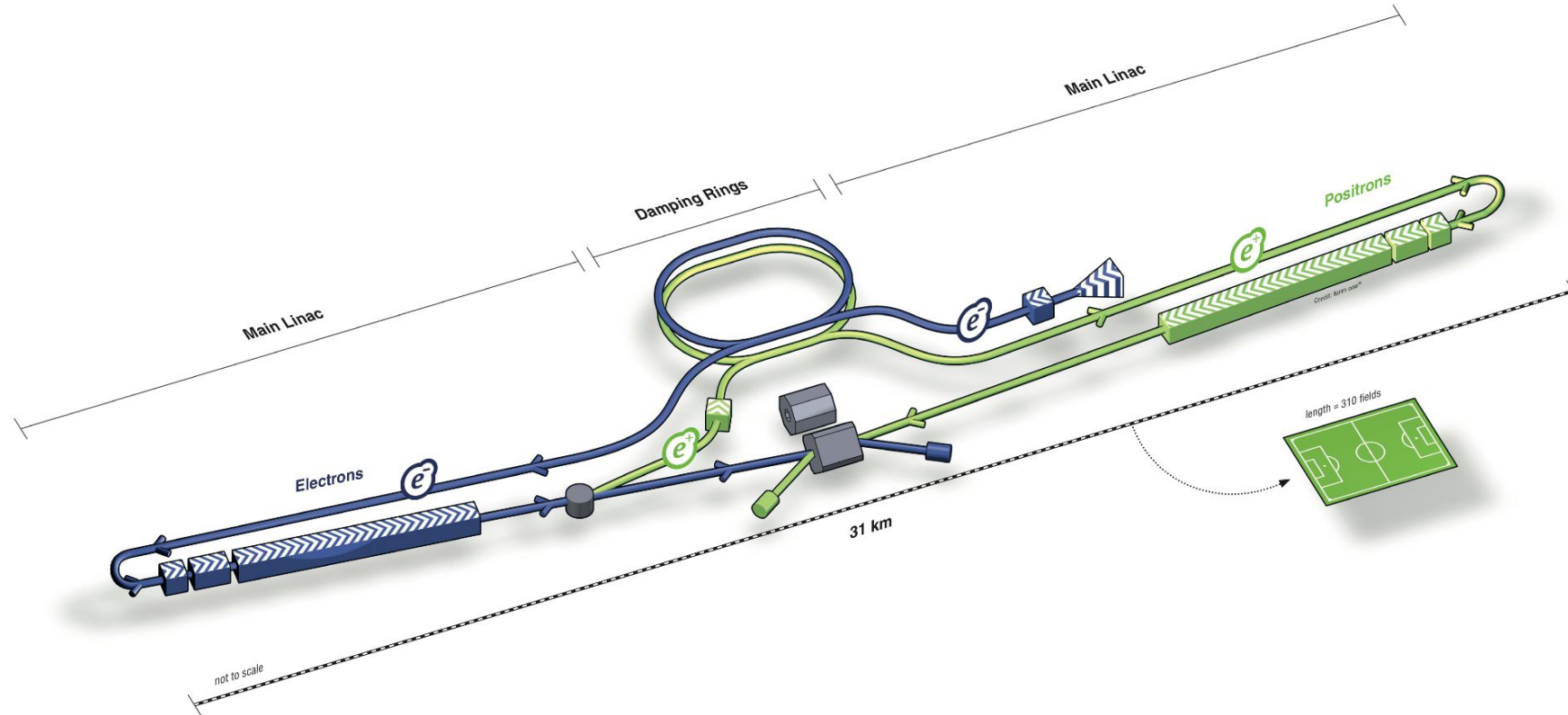




A black and white artistic rendering of a particle accelerator facility. Two bright, glowing beams of light represent the particle paths, extending from the foreground into the distance. The beams are flanked by circular cross-sections showing the internal structure of the accelerator. The background features a vast, flat landscape with distant mountains under a dark sky filled with stars.

Positron Acceleration and Future Colliders

The Linear Collider



Linear colliders are large machines that collide electron and positron beams.

Plasma wakefield acceleration has the potential to dramatically reduce the size and cost of the machine

10 TeV Wakefield Collider Design Study

Challenges

Energy Recovery

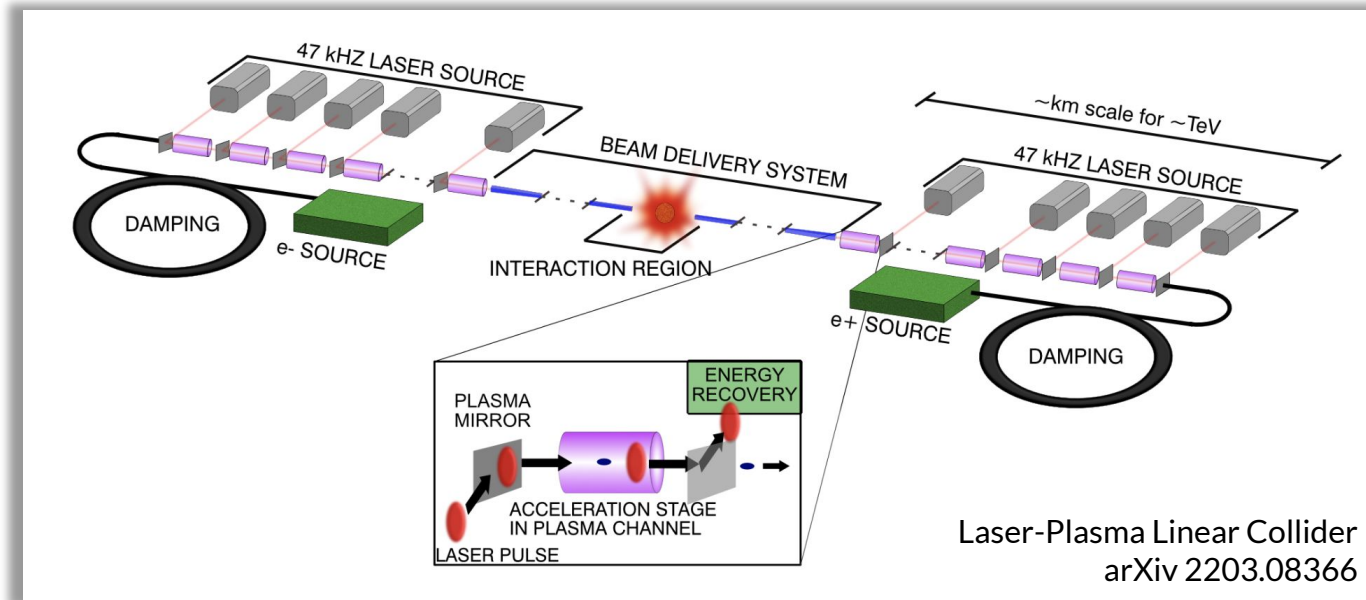
Beam Quality

Repetition Rate

Positron Acceleration

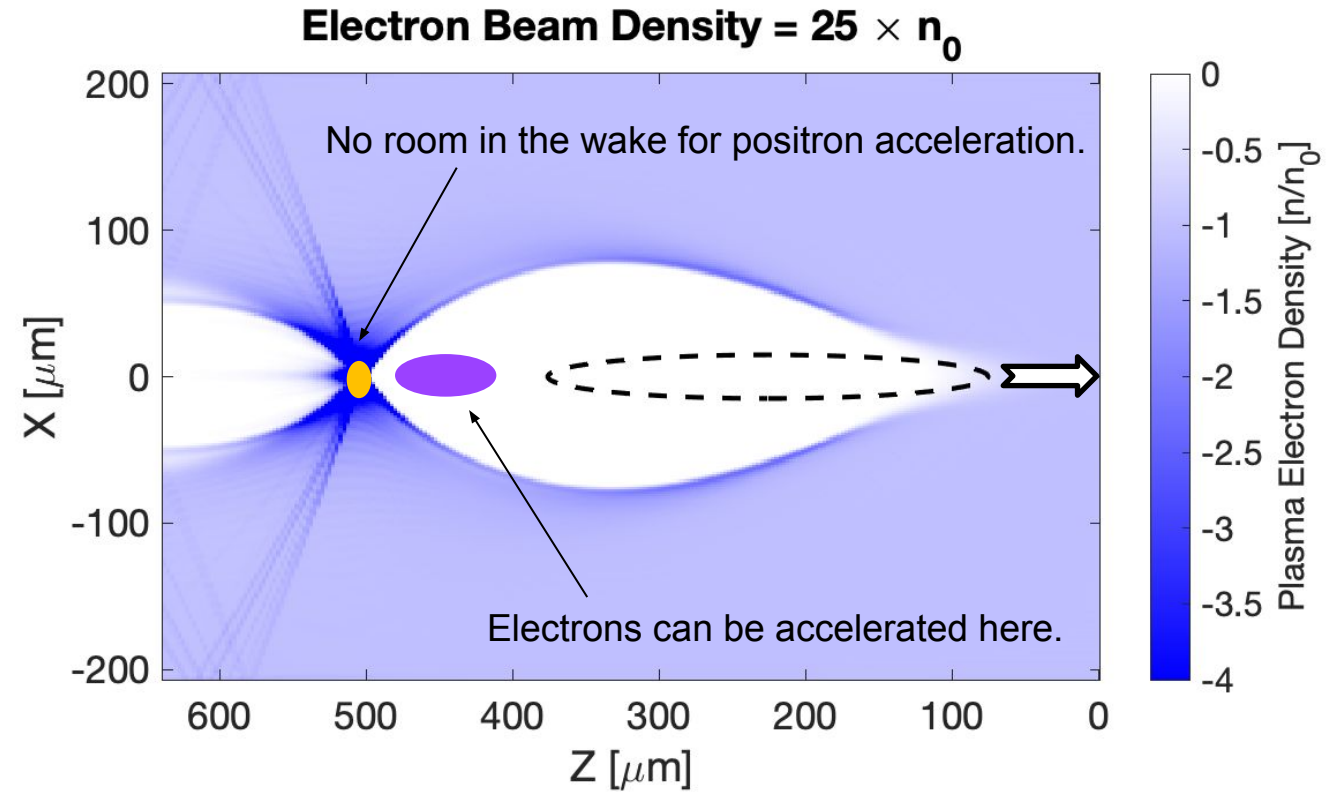
Staging

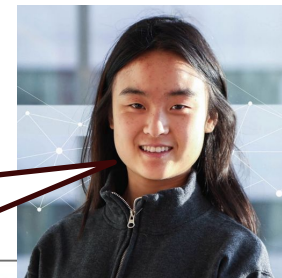
Beam-Beam Physics



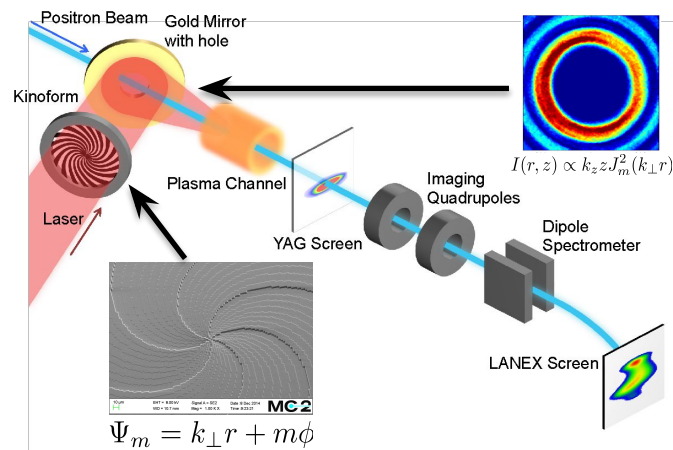
Positron acceleration?

There is no part of the nonlinear wakefield that is *uniformly* accelerating and focusing for positrons.

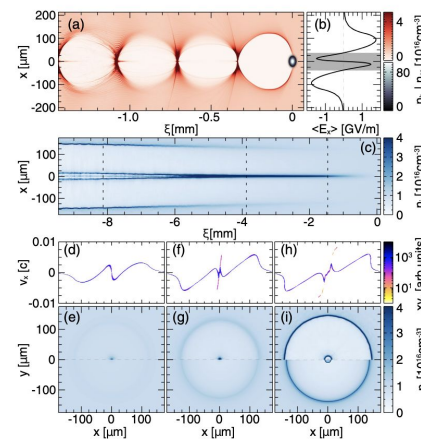




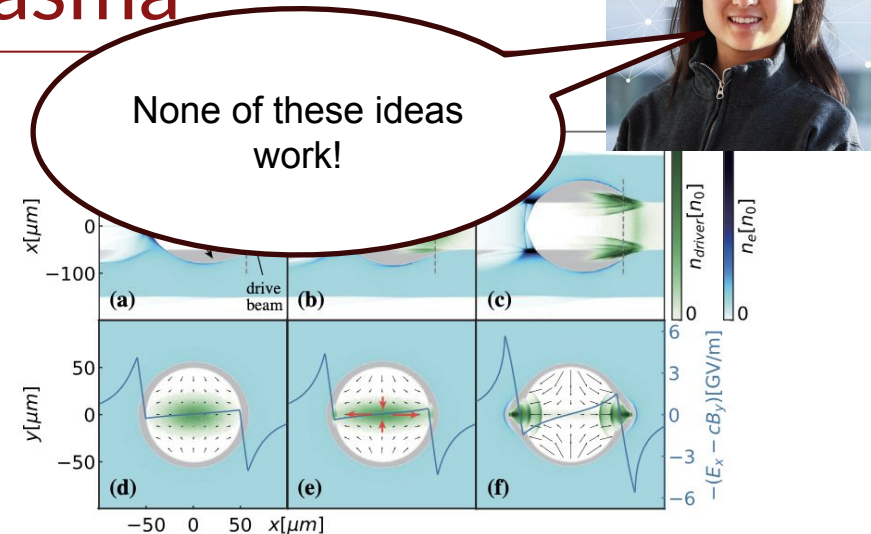
Concepts for Positron Acceleration in Plasma



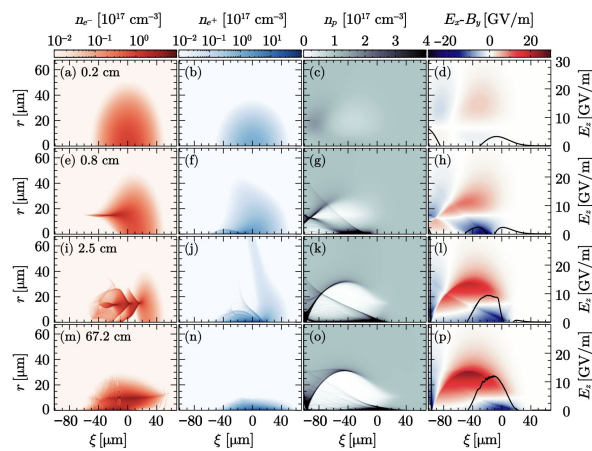
Hollow Channel Plasma
Gessner et al. *Nat. Comm.* 7, 11785 (2016)



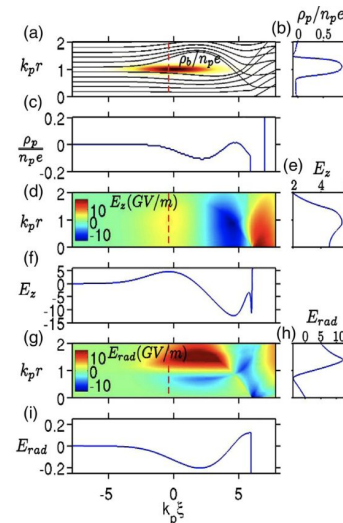
Warm Plasma Channels
Silva et al. *Phys. Rev. Lett.* 127, 104801 (2021)



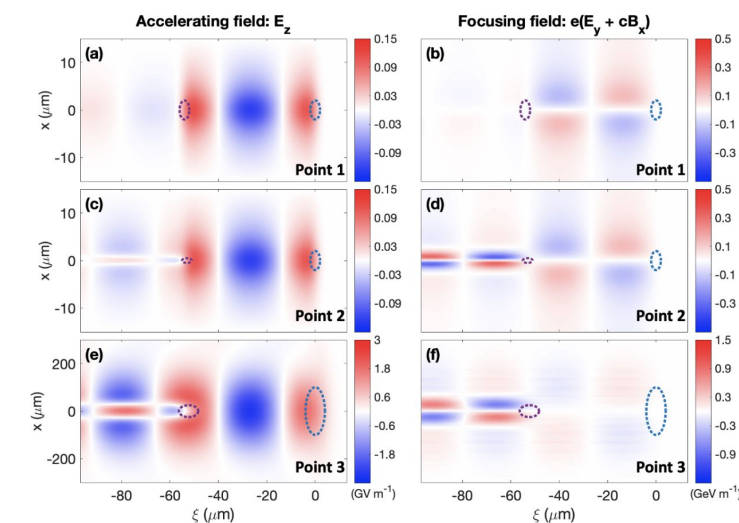
Asymmetric Plasma Channels
Zhou et al. *Phys. Rev. Lett.* 127, 174801 (2021)



Non-neutral Fireball Acceleration
Silva et al. *Phys. Rev. Accel. Beams* 26, 091301 (2023)

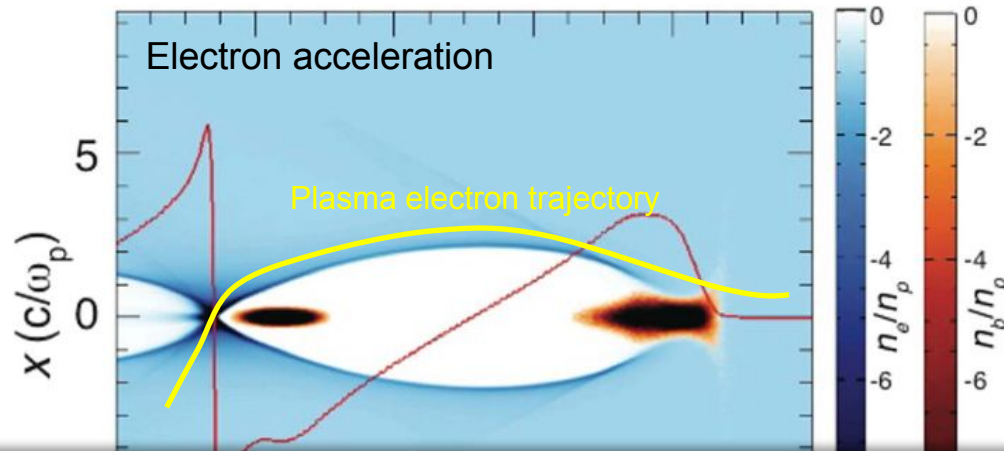


Donut-Shaped Drivers
Jain et al. *Phys. Rev. Lett.* 115, 195001 (2015)



Quasi-Linear Plasmas
Hue et al. *Phys. Rev. Research* 3, 043063 (2021)

Can we understand the issue?



PRL 95, 195002 (2005)

PHYSICAL REVIEW LETTERS

week ending
4 NOVEMBER 2005

Effects of Ion Motion in Intense Beam-Driven Plasma Wakefield Accelerators

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¹Department of Physics and Astronomy, University of California Los Angeles, Los Angeles, California 90095, USA

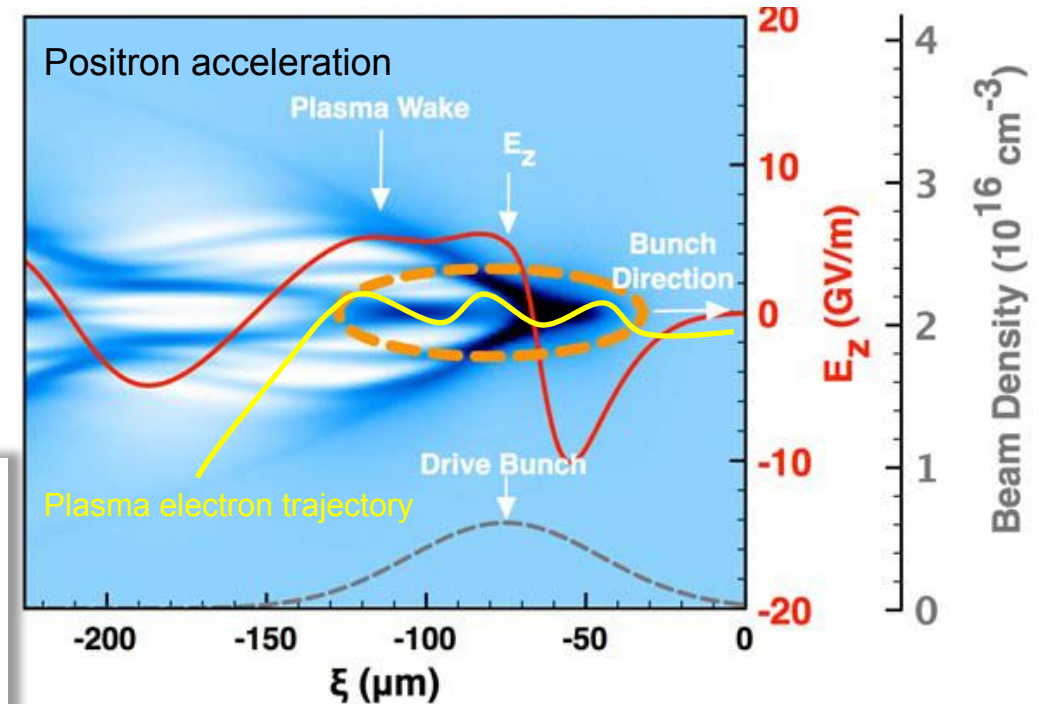
²Department of Physics, University of California Santa Barbara, Santa Barbara, California 93016, USA

³Department of Physics, Manhattan College, 4513 Manhattan College Parkway, Riverdale, New York 10471, USA
(Received 27 June 2005; published 31 October 2005)

Recent proposals for using plasma wakefield accelerators (PWFA) as a component of a linear collider have included intense electron beams with densities many times in excess of the plasma density. The beam's electric fields expel the plasma electrons from the beam path to many beam radii in this regime. We analyze here the motion of plasma ions under the beam fields, and find for a proposed PWFA collider scenario that the ions completely collapse inside of the beam. Simulations of ion collapse are presented. Implications of ion motion on the feasibility of the PWFA-based colliders are discussed.

DOI: 10.1103/PhysRevLett.95.195002

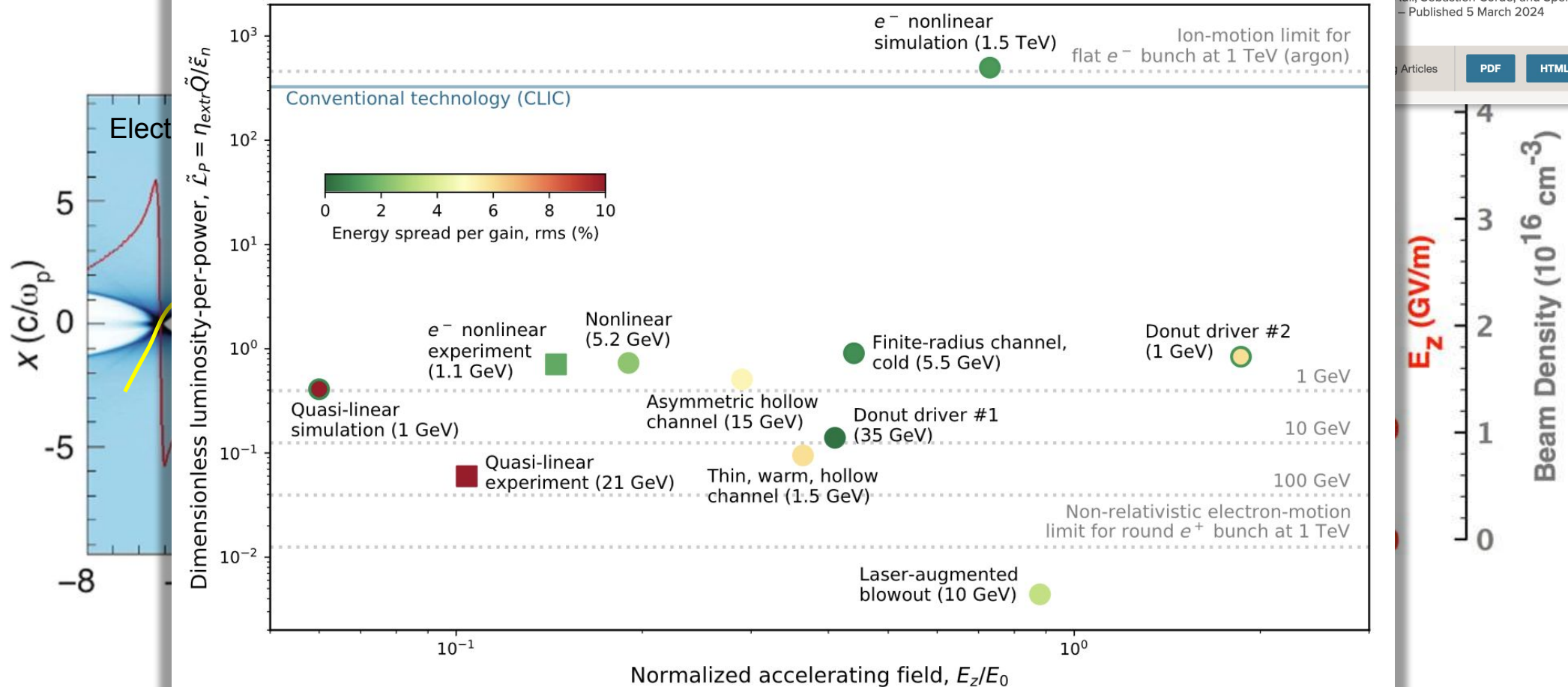
PACS numbers: 52.40.Mj, 29.17.+w, 29.27.-a, 52.75.Di



$$\frac{d^2 r}{d\xi^2} = -\frac{Zr_a N_b}{A\epsilon_{n,x}\sigma_z} \sqrt{r_e n_0 \gamma} r = -k_i^2 r, \quad (5)$$

$$\Delta\phi \simeq k_i \Delta\xi = \sqrt{\frac{2\pi Zr_a \sigma_z N_b}{A\epsilon_{n,x}}} (r_e n_0 \gamma)^{1/4}. \quad (6)$$

Can we understand the issue?

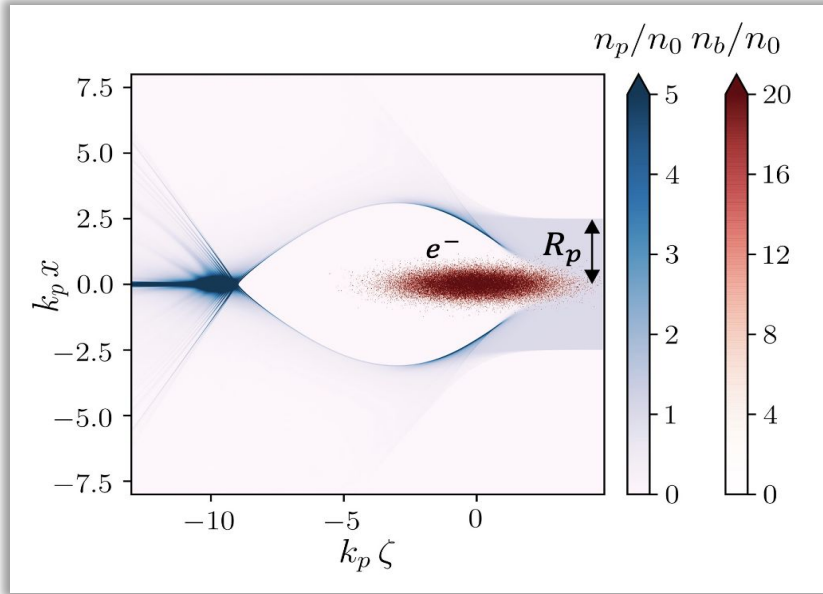


The “electron motion challenge”:

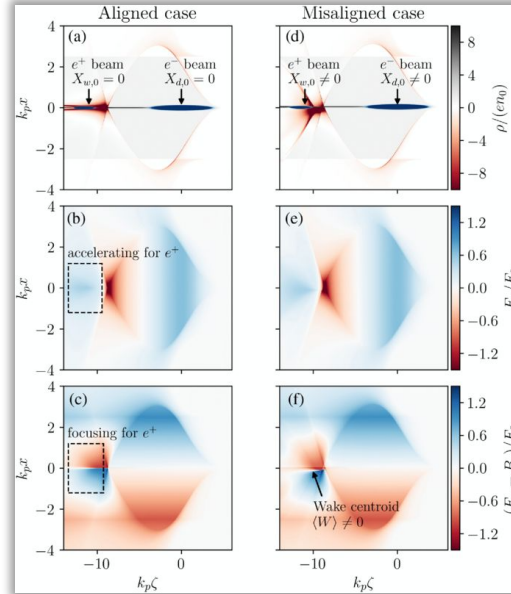
$$\Delta\phi_e \simeq k_e \Delta\zeta = \sqrt{\frac{\mu_0 e^2}{2} \frac{\sigma_z N}{\gamma_{pe} m_e}} \sqrt{\frac{r_e \gamma \Delta n}{\epsilon_{nx} \epsilon_{ny}}}$$

New Direction: Filament Plasmas

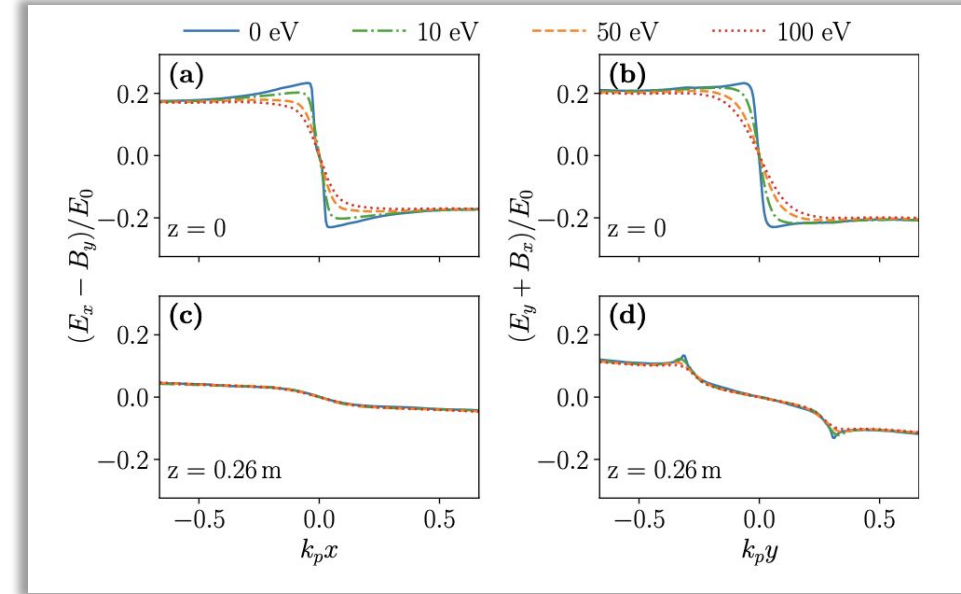
S. Diederichs et al, PRAB (2019)



S. Diederichs et al, PRAB (2022)

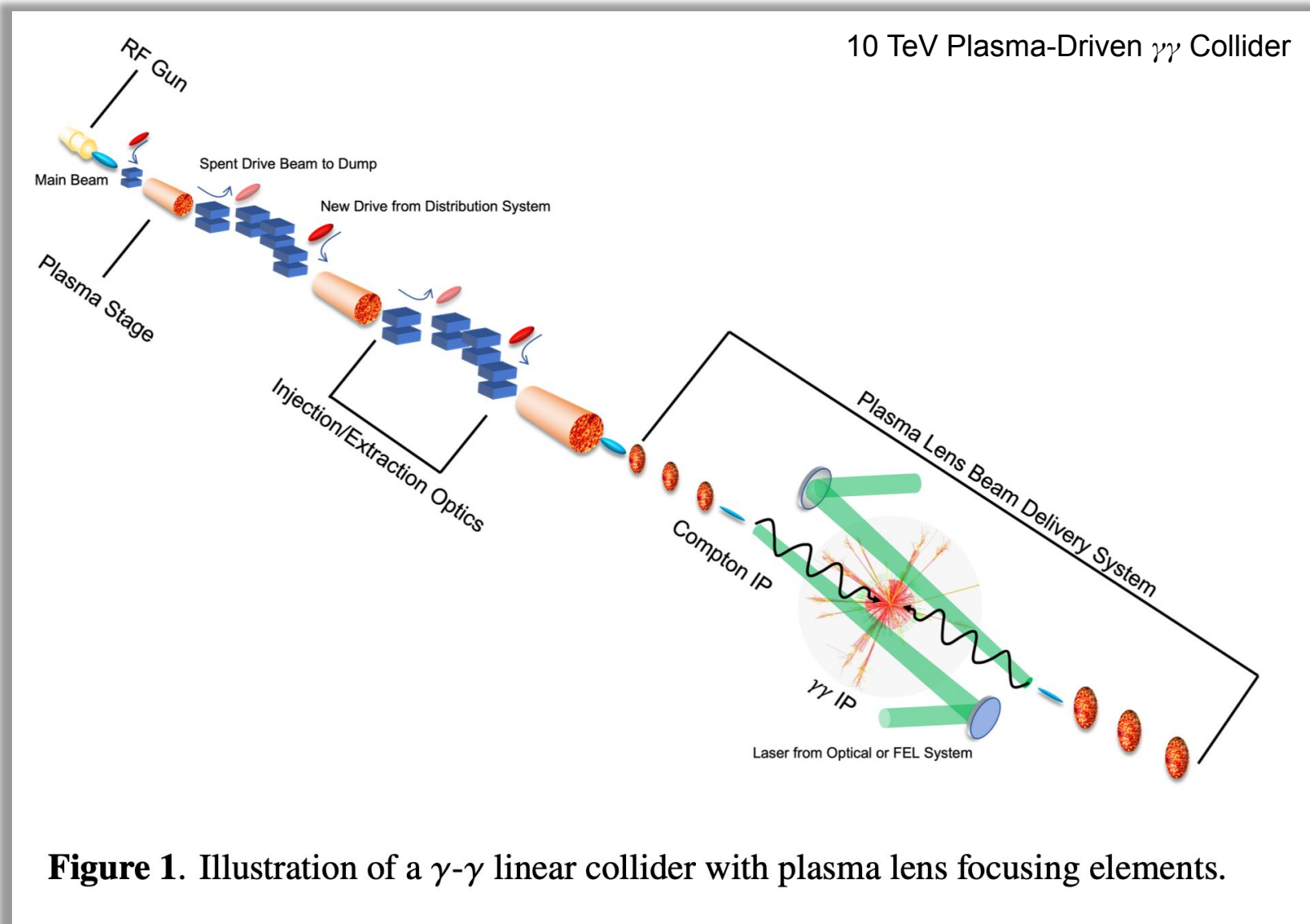


S. Diederichs et al, PoP (2023)



The E333 experiment at FACET-II aims to demonstrate electron beam-driven wakes in filament plasmas for stable, high-quality positron acceleration.

Gamma-Gamma Collider





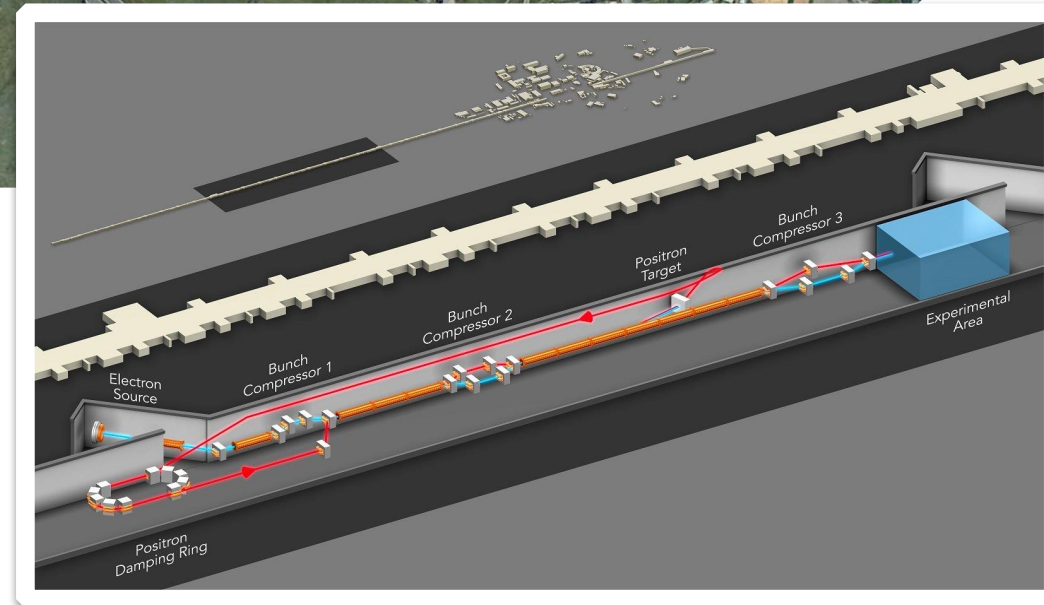
What's Happening Now?

FACET-II: A world-class test beam facility



What makes FACET-II unique?

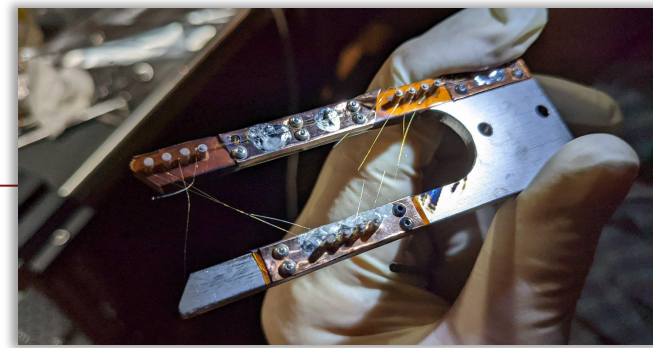
- Ultra-short bunches with ultra-high peak current.
- Precise control of the longitudinal bunch profile.
- High-power laser and plasma sources.



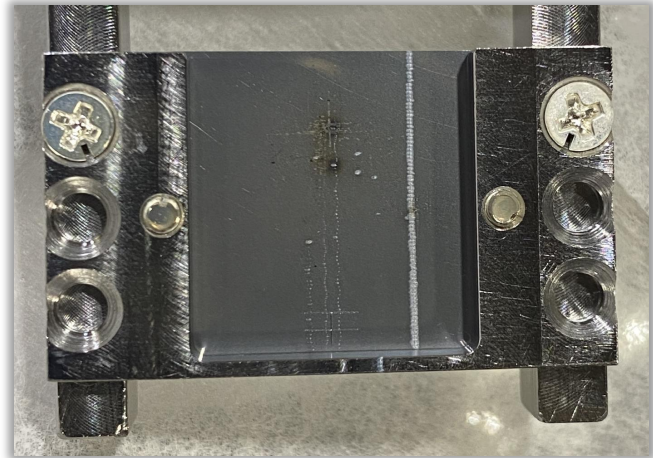
FACET-II: Extreme Beams

Ultra-short, ultra-high peak current bunches produce extreme fields. The beam eats intercepting diagnostics for breakfast!

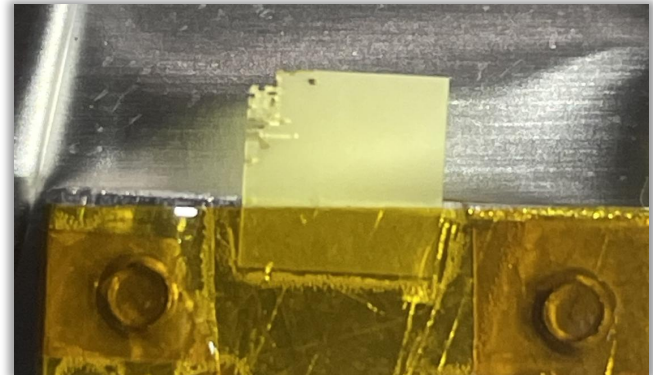
The same fields that destroy diagnostics are exactly what we want for driving high-amplitude wakes in plasma!



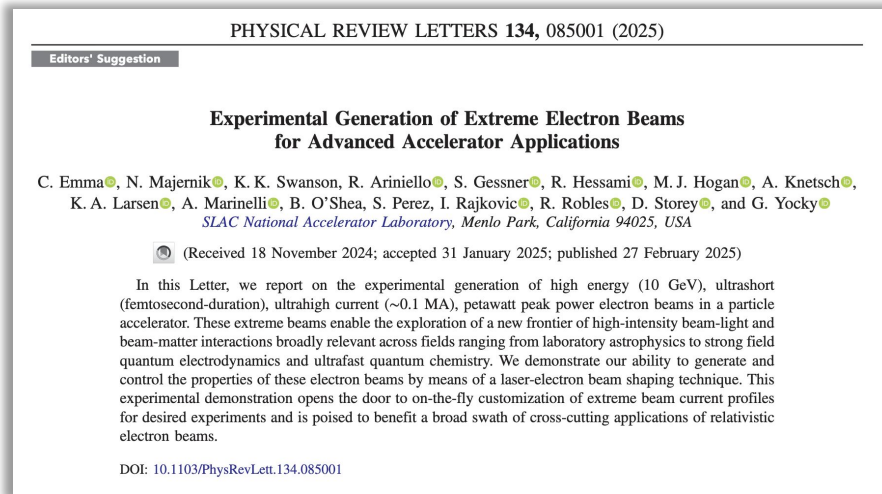
Wire scanner. . . shredded



OTR foil. . . drilled



YAG crystal. . . swiss-cheesed



HOME > NEWS > ALL NEWS > FIRST PETAWATT ELECTRON BEAM ARRIVES, READY TO RIP APART MATTER AND SPACE

NEWS | PHYSICS

First petawatt electron beam arrives, ready to rip apart matter and space

Ultra-high-power particle pulses could boost x-ray science and laboratory astrophysics

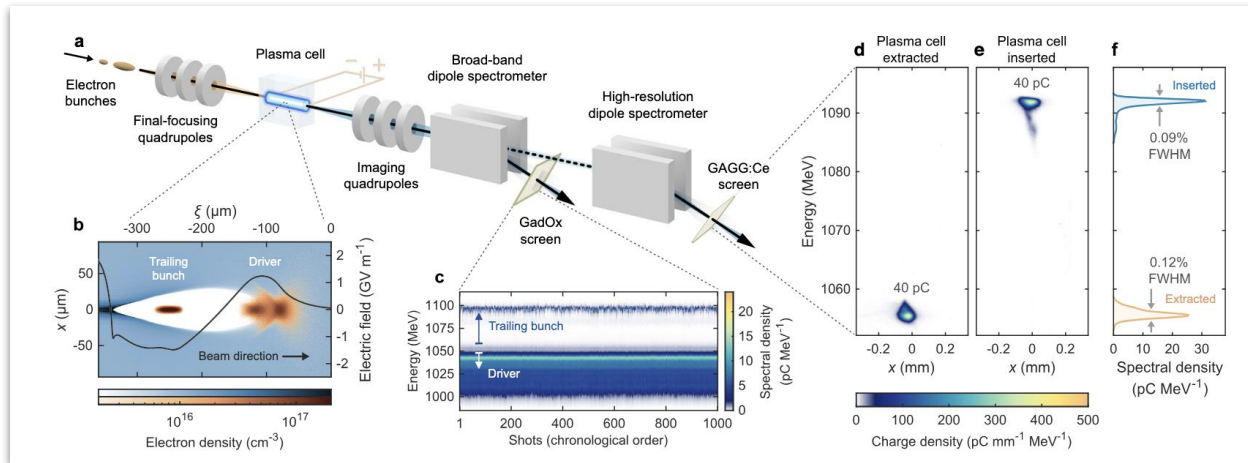
27 FEB 2025 • 5:05 PM ET • BY [ADRIAN CHO](#)



In this experimental demonstration, researchers control the properties of these electron beams by means of a laser-electron beam shaping technique. This experimental demonstration opens the door to on-the-fly customization of extreme beam current profiles for desired experiments and is poised to benefit a broad swath of cross-cutting applications of relativistic electron beams.

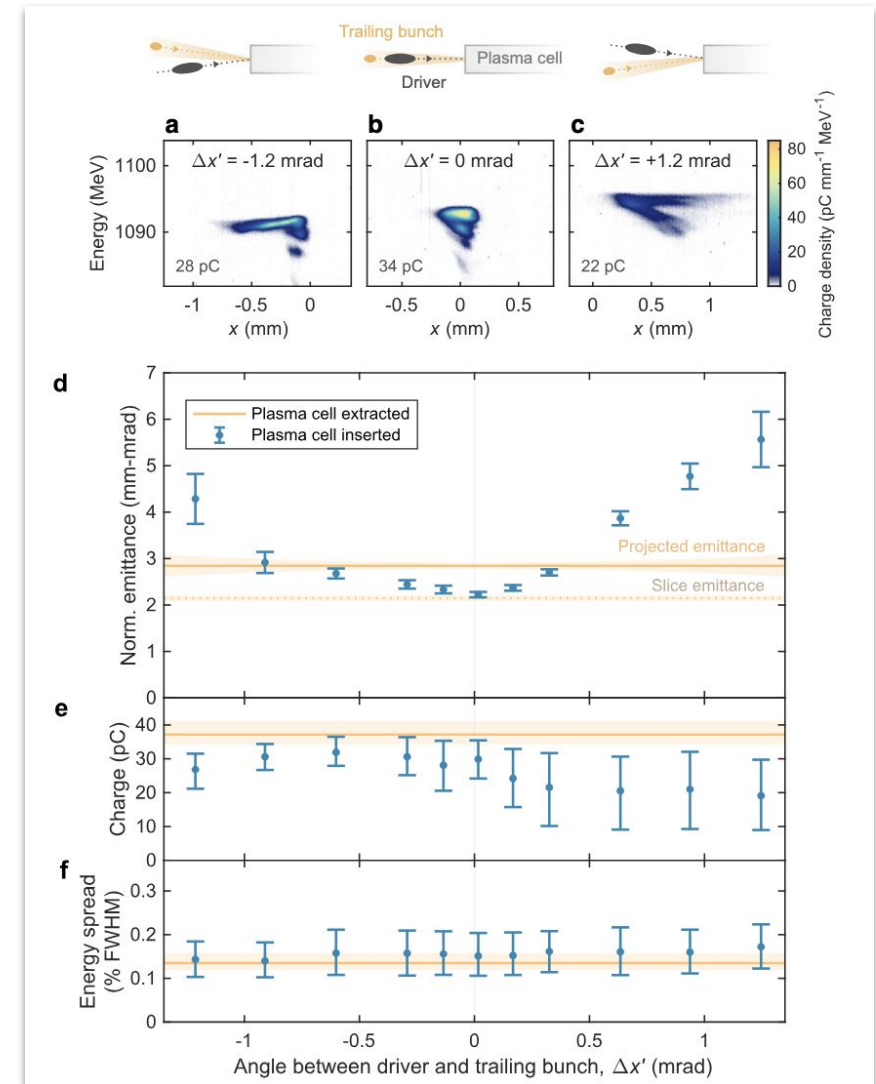
DOI: 10.1103/PhysRevLett.134.085001

Progress in controlling plasma acceleration



Emittance preservation in a plasma-wakefield accelerator,
C. Lindstrøm et al, Nature Communications (2024)

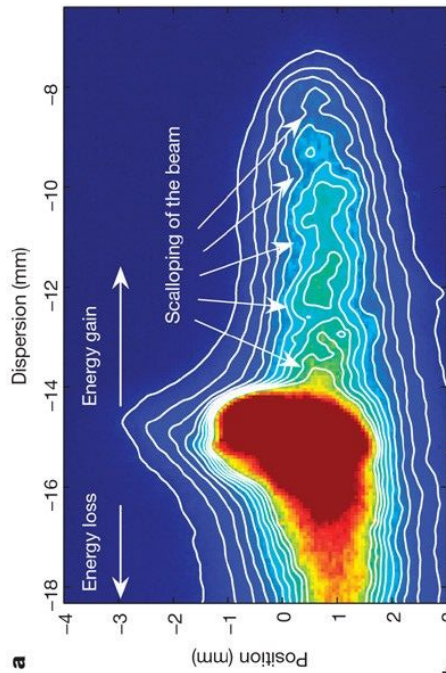
First demonstration of emittance
preservation in a plasma accelerator!



Progress in controlling plasma acceleration

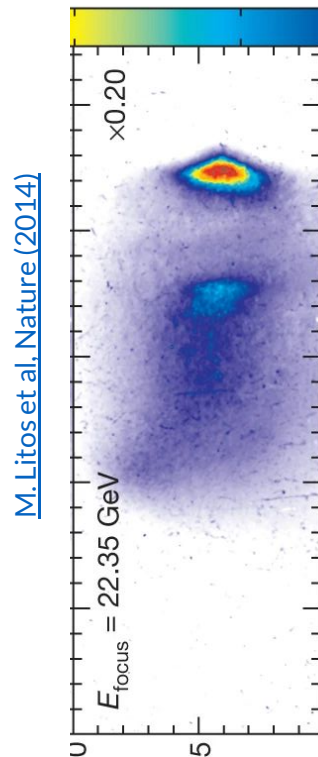
SLAC FFTB 2006

$$E_{\text{gain}} = 42 \text{ GeV}$$
$$\sigma_E = 100\%$$



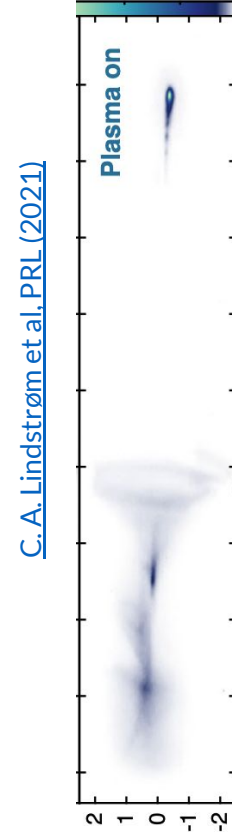
SLAC FACET 2014

$$E_{\text{gain}} = 2 \text{ GeV}$$
$$\sigma_E = 2\%$$



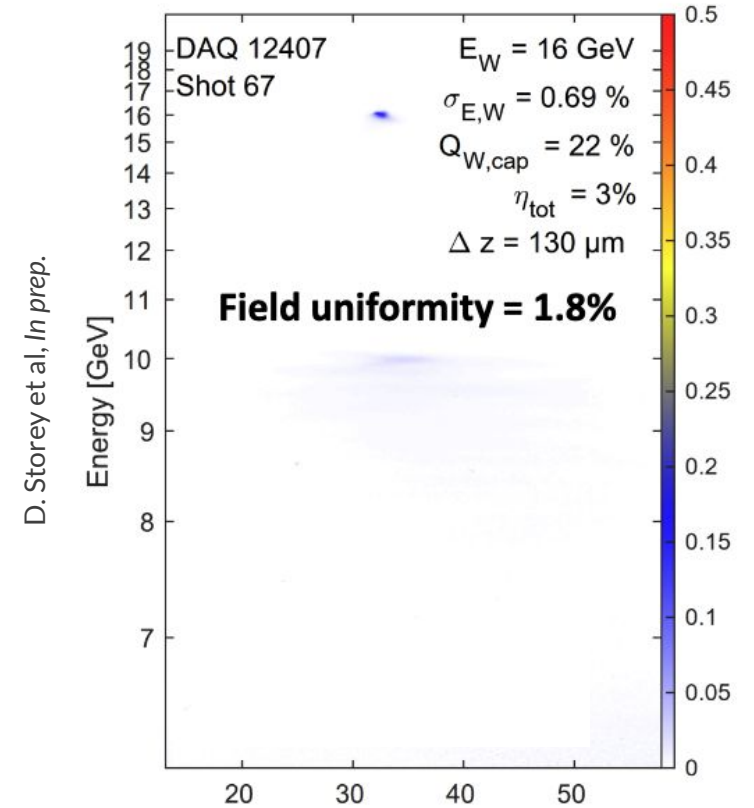
DESY FlashForward 2021

$$E_{\text{gain}} = 40 \text{ MeV}$$
$$\sigma_E = 0.13\%$$



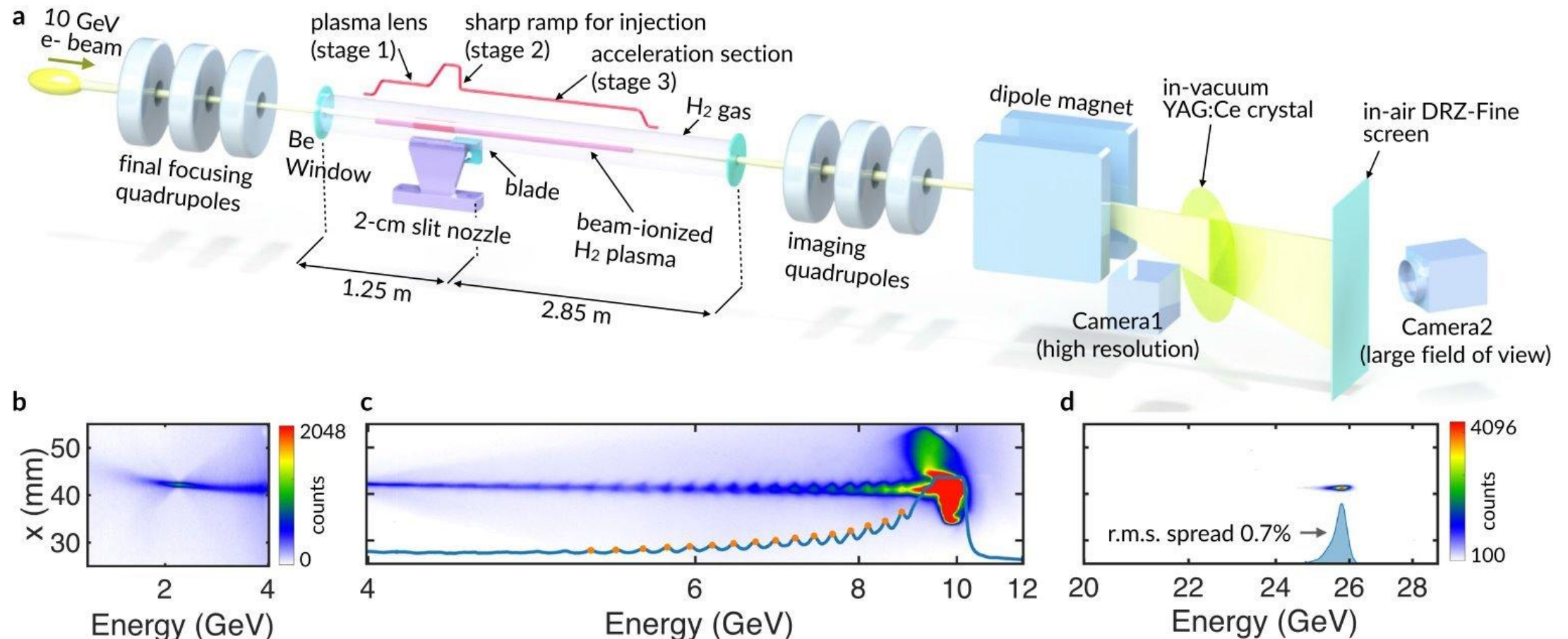
SLAC FACET-II 2025

$$E_{\text{gain}} = 6 \text{ GeV}$$
$$\sigma_E = 0.69\%$$



Improvements in Beam Test Facilities have yielded quantitative progress toward collider-quality beams from beam-driven plasma acceleration.

High Quality Beams from Plasma



Enabling new light source applications

Conclusion

Plasma wakefield acceleration is a demonstrated technique for accelerating particles to ultra-high energies in ultra-short distances.

Rapid progress on addressing key challenges related to beam quality from plasma accelerators.

But there is much work left to be done!

Unfortunately, support for plasma accelerator R&D will decrease in the US, at least for a few years.

Nevertheless, students continue to produce high-impact results by engaging in PWFA research!